Intelligence

Lecture 8

Heritability of Intelligence

Dr Steven Poulter

Key Themes

- Looking at twin studies
- Looking at DNA
- Stability of IQ across the lifespan and generations

Glossary of Key Terms

Genes
Segments of DNA that code for proteins and influence physical and psychological traits, including aspects of intelligence.

Proteins
Molecules produced from gene instructions that carry out most biological functions, including those in the brain.

Cognitive Power
A general term for mental abilities such as reasoning, memory, problem-solving, learning, and processing speed.

Heritability
The proportion of differences between individuals in a trait (e.g., intelligence) that can be explained by genetic variation within a population.

Shared Environment
Environmental influences that people raised together tend to share, such as household experiences, socioeconomic conditions, or parenting style.

Non-shared Environment
Environmental factors unique to each individual - experiences, peers, illnesses, teachers - making people less similar to their siblings.

Monozygotic (MZ) Twins
Identical twins who share 100% of their genes; used to estimate genetic influences on traits.

Dizygotic (DZ) Twins
Non-identical twins who share roughly 50% of their genes; used as a comparison group in behavioural genetics.

Twin Studies
A major method for estimating the contributions of genes, shared environment, and non-shared environment by comparing similarities between MZ and DZ twins.

Family Studies
Research exploring how traits run in biological families to estimate genetic influence.

Adoption Studies
Studies comparing adopted children to their biological relatives (genetic influence) and adoptive families (environmental influence).

Correlation Coefficient
A numerical estimate of similarity between individuals on a trait. Higher correlations for MZ than DZ twins suggest genetic influence.

Genome-Wide Association Study (GWAS)
A genetic method scanning the whole genome to identify variants associated with traits such as intelligence.

Single Nucleotide Polymorphism (SNP)
A single DNA base variation. Individuals differ at millions of these positions, making them useful markers in GWAS.

General Intelligence (g)
A statistical construct reflecting the shared variance across different cognitive tasks; commonly used in genetic and psychological research.

Gene × Environment Interaction
The concept that genes and environments do not act separately but influence each other - environmental experiences can activate or modify genetic tendencies.

Genetic Contribution
The proportion of variation in intelligence explained by genetic differences between people.

Shared Environmental Contribution
The proportion of variation explained by factors siblings share, typically decreasing as people get older.

Non-shared Environmental Contribution
The portion of variation explained by unique individual experiences, relatively stable across development.

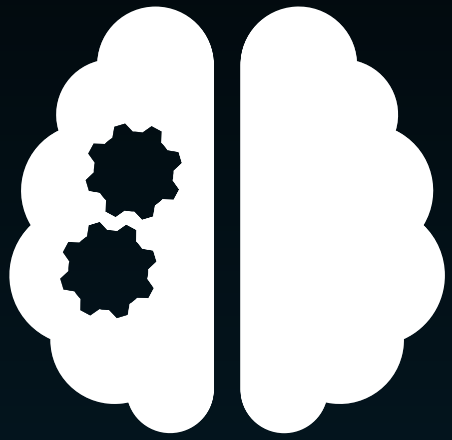
Stability of Intelligence
The observation that intelligence remains relatively consistent throughout adulthood, even though small changes can occur across the lifespan.

Flynn Effect
The sustained increase in average IQ scores across generations, particularly pronounced for non-verbal (fluid) intelligence tests.

What are genes for?

Genes

Proteins

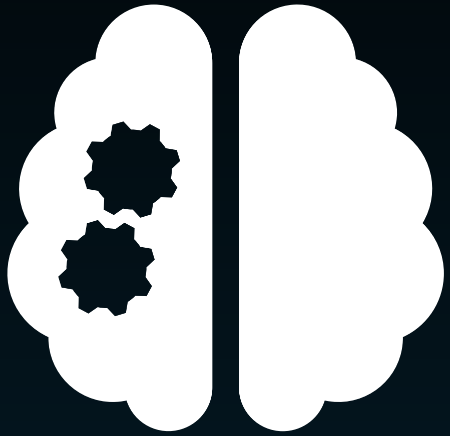


Cognitive Power

What are genes for?

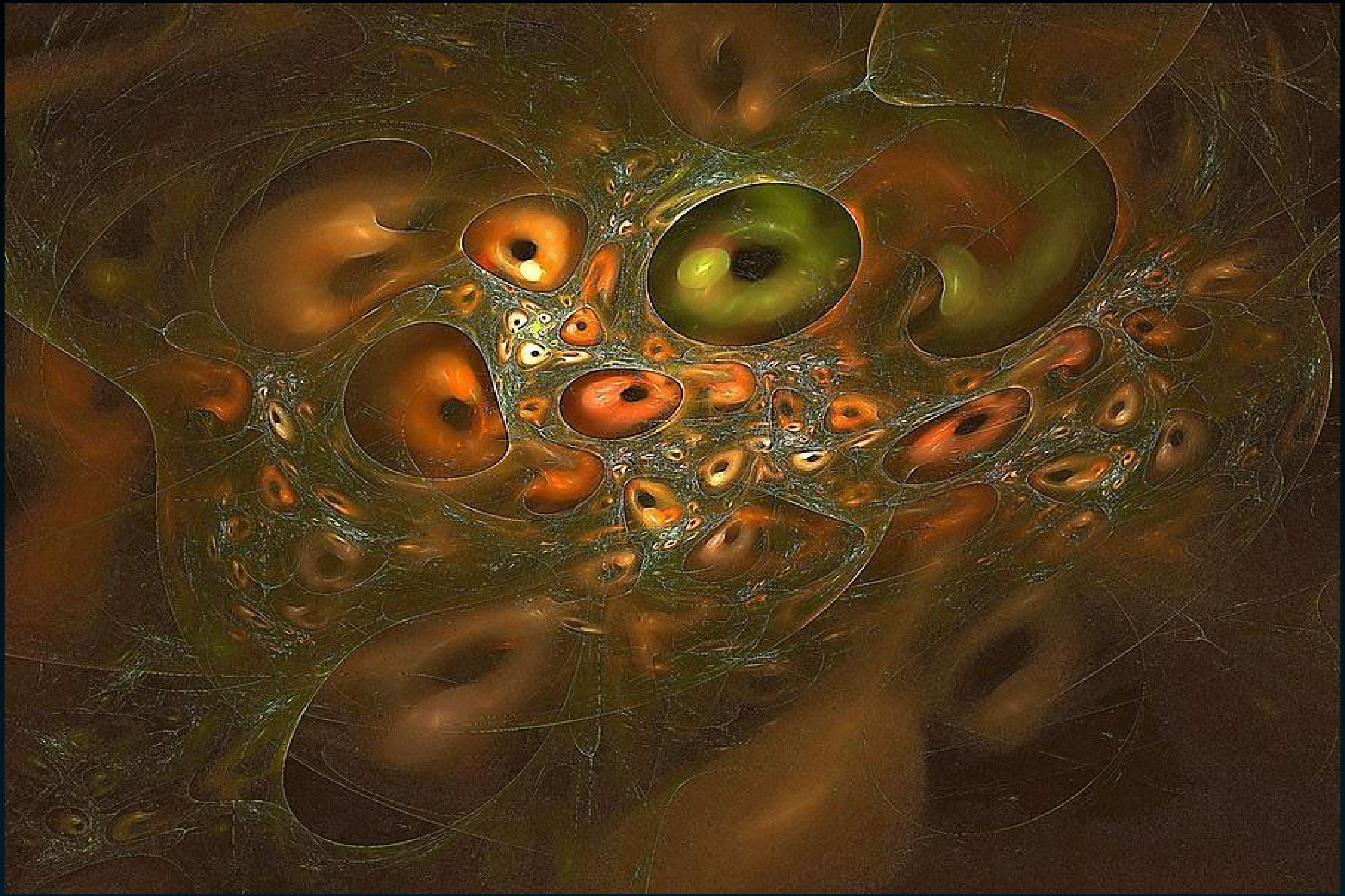
Genes

Whole lot going on
in between



Cognitive
Power

Where did genes come from?



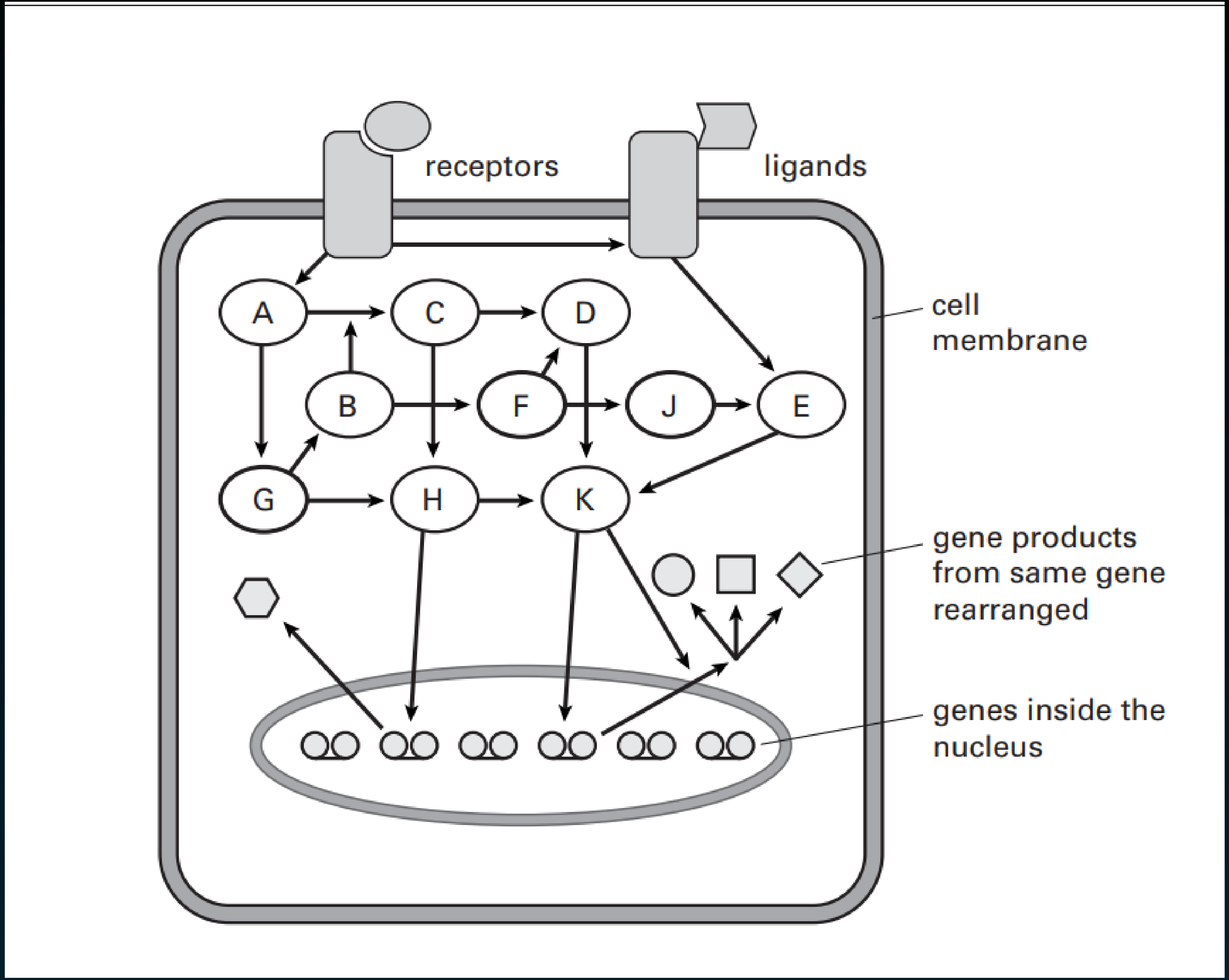
Where did order and complexity in living things originate?

Where did genes come from?

Living beings operate under rapid environmental change

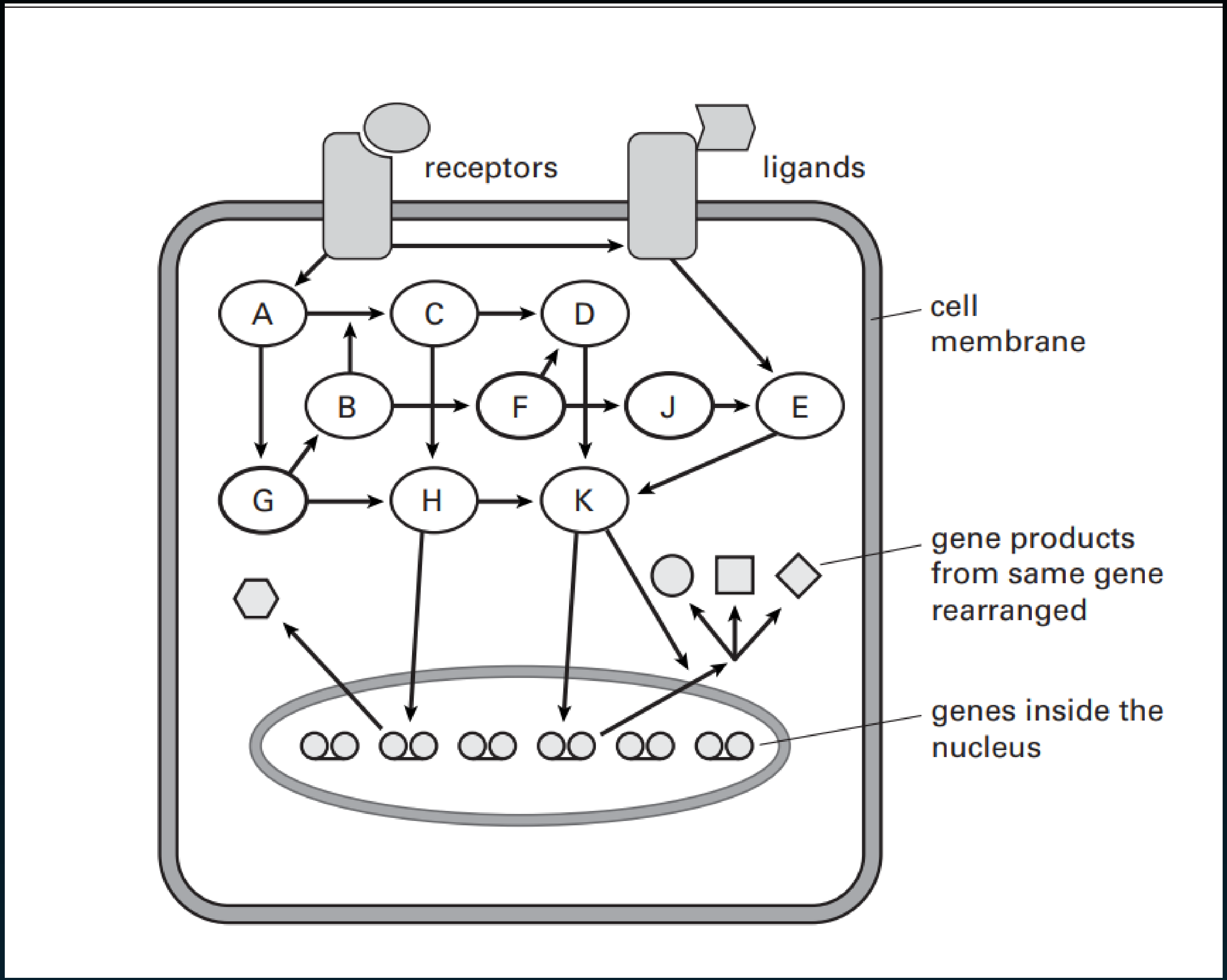
This reflects the self-organising properties of the system

Do cells think?

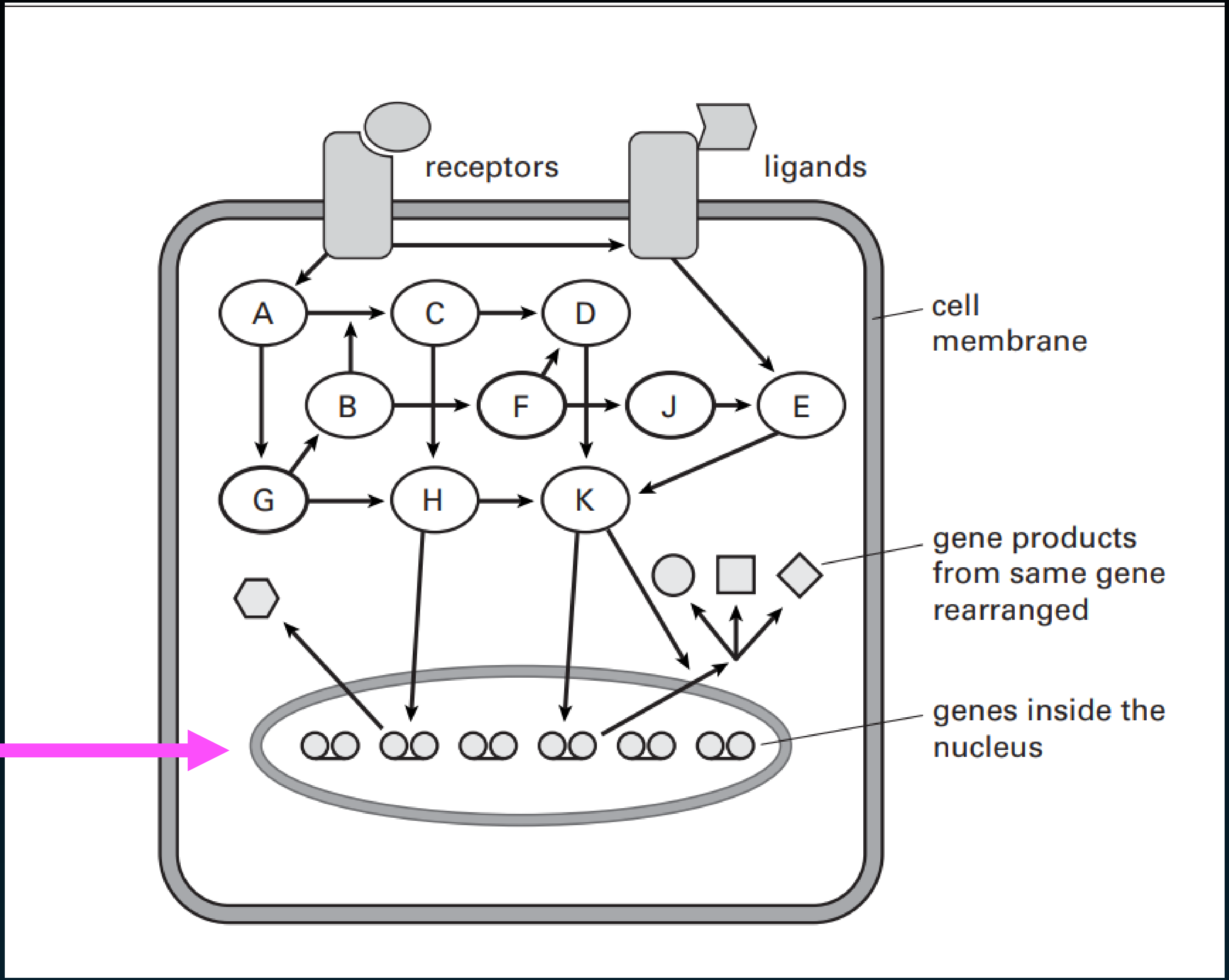


Do cells think?

Fixed DNA codes cannot register the changing statistical patterns needed to survive in rapidly changing environments.

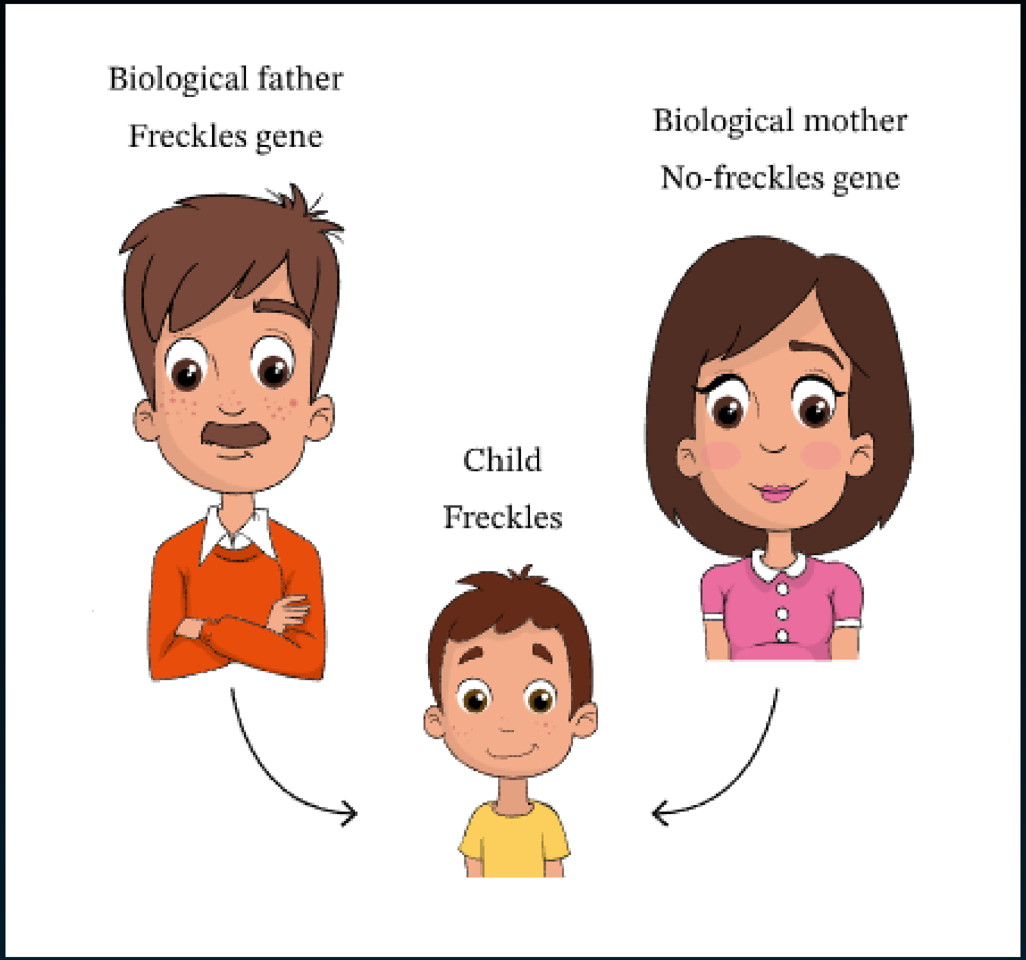


Do cells think?



End point for many of these signalling pathways will be gene transcriptions

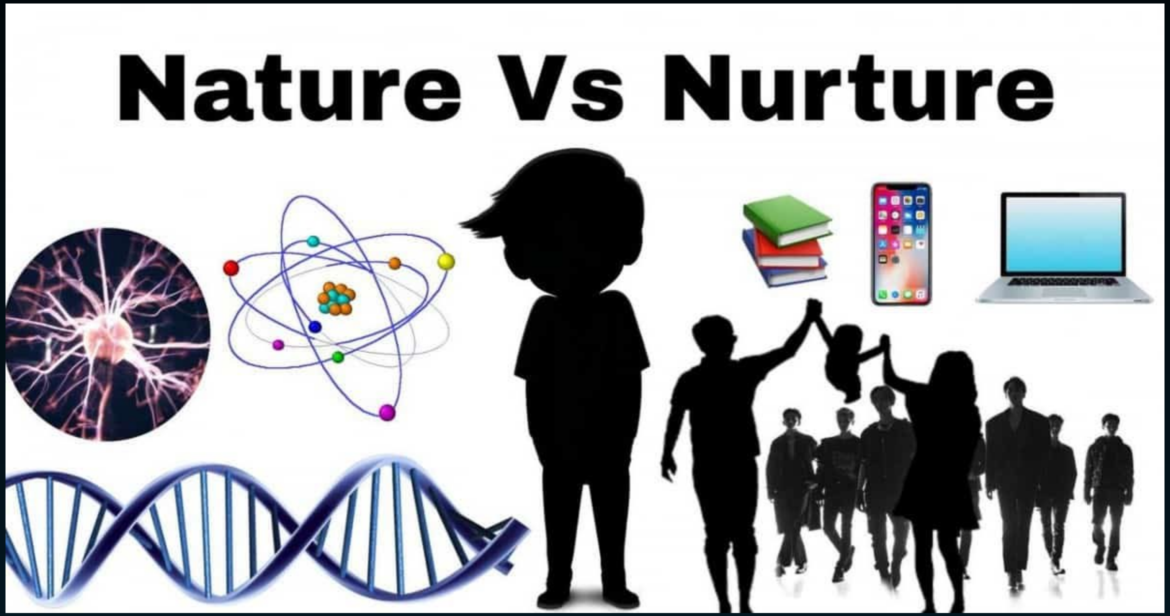
What do we mean by heritability?



The heritability of human **physical characteristics**, such as having freckles or a nose is genetic in origin and not influenced by environment

The proportion of shared variance is high, e.g. 100%

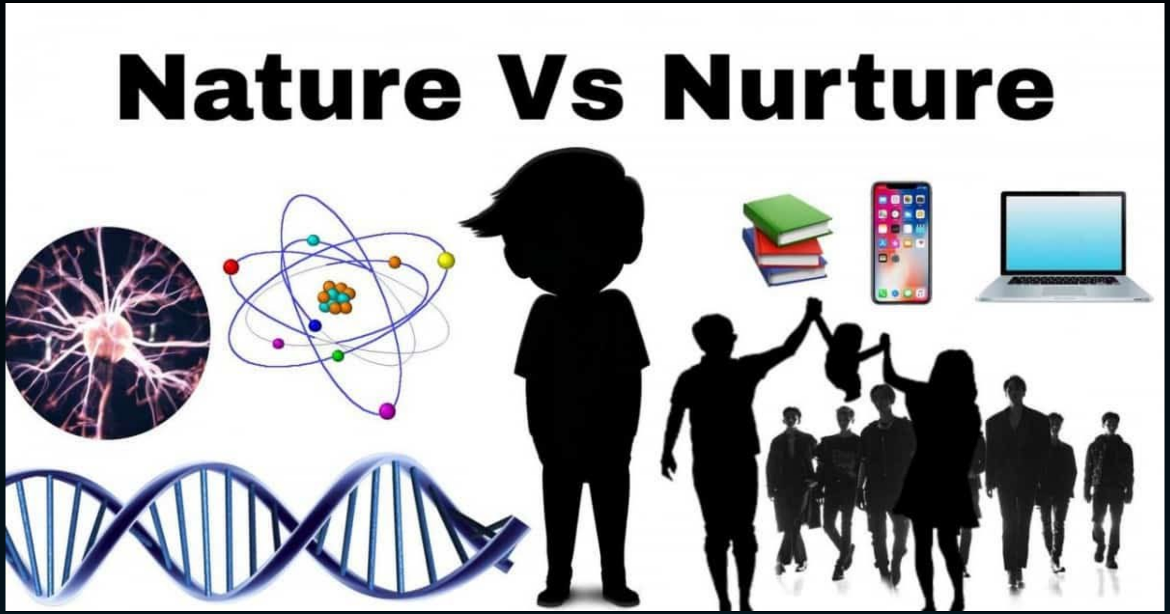
What do we mean by heritability?



However, some aspects of human behaviour, such as intelligence can be highly influenced by environment as well as genetic

The proportion of shared variance is lower, e.g. 50%

What do we mean by heritability?



One method to assess the heritability of intelligence is to **estimate** the relative strength of genes vs environment.

The influence of genetic and environmental components in this theory account for **100%** of the variance

If a study found genetics accounted for 30% variance for intelligence, we would estimate that environment accounts for 70% of the variance of intelligence

However….simply adding together genetic and environmental factors is too simplistic

- Intelligence should not be seen as the result of ‘genetics + environment’ but rather of **‘genetics × environment’**

- Better view: relative influences of genes and environment on intelligence is the result of a long-term interaction, with environmental factors triggering certain genetic behaviours and the effects of the environment differing between individuals because of their genetic makeup.

Methods for assessing heritability?

Family Studies

Adoption Studies

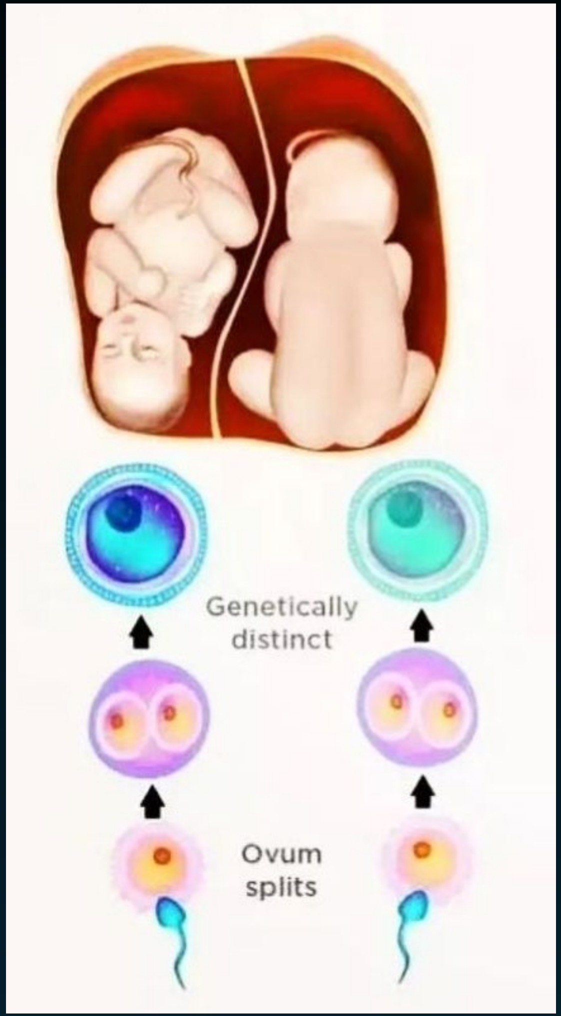
Twin Studies

TWIN STUDIES

Monozygotic twins (identical twins)

Dizygotic twins (non-identical twins)

Same fertilised egg and same genetic make-up



Two different eggs and two different sperm. Share half genetic make-up

Monozygotic twin (identical twins) vs dizygotic twin (non-identical twins) studies

- Comparing different twins provides natural experiment
- Can compare people of the same age
- In some cases they have same genetic make-up and in some they share half

To what extent are differences in intelligence caused by differences in genes versus differences in environment?

Monozygotic twin (identical twins) vs dizygotic twin (non-identical twins) studies

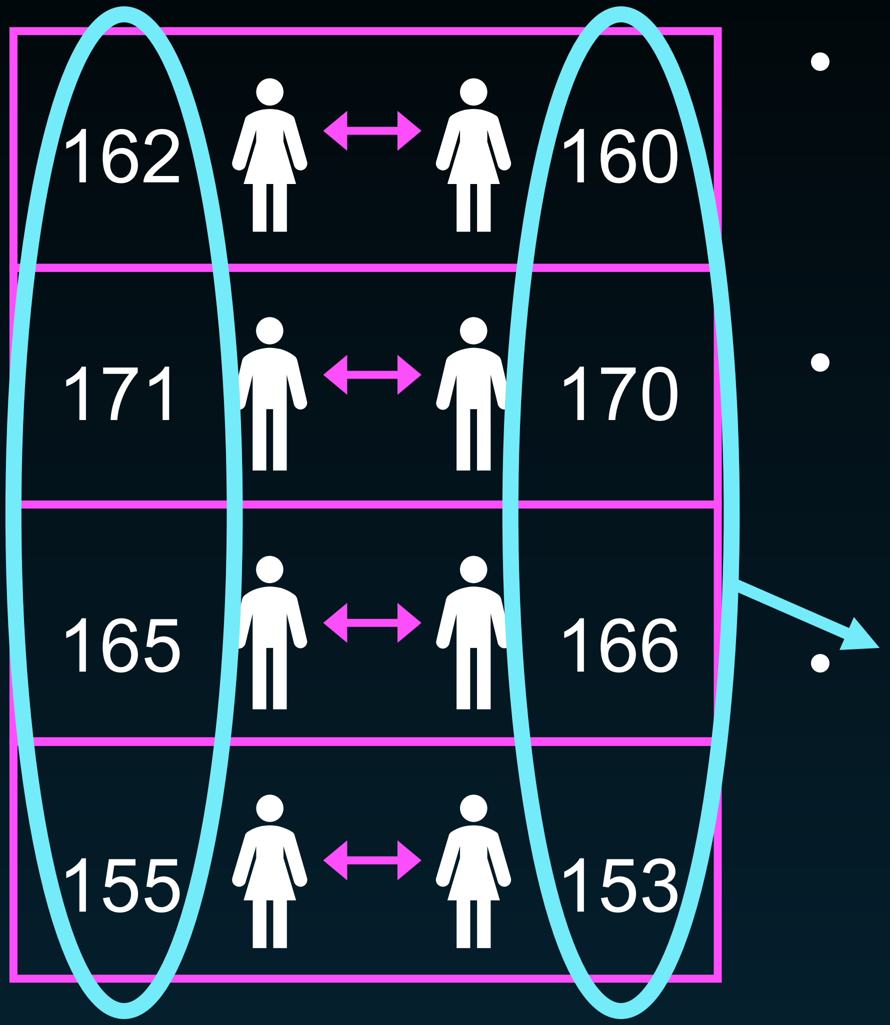
- Common in twin studies to divide the environment into:

  > Shared (common to both)
  > Non-shared (unique to each)

- Effect of genes on measurable traits (height, extraversion, intelligence etc.)

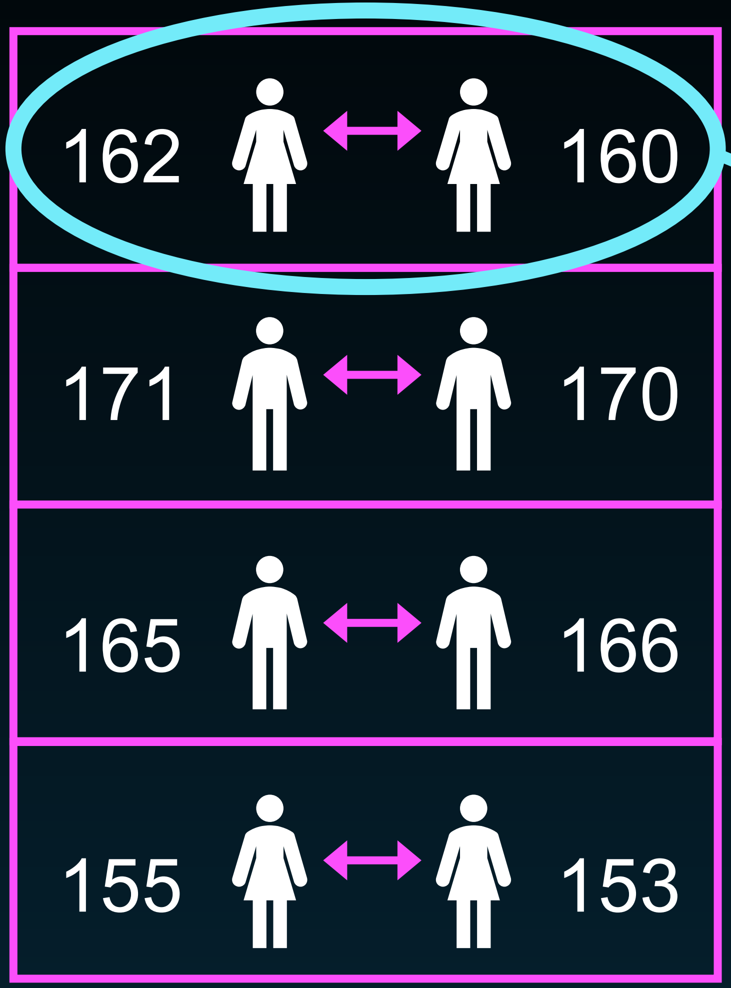
Contributions of genes, shared environment, and non-shared environment to intelligence differences

Monozygotic twins
(identical twins)



- A number for each member of the twin pair is calculated. For this example it is height in cm
- Between-twin correlation to see if when one member of a twin pair has a high score on a trait, the other member also scores highly
- When we compare the columns derived from each twin, we can calculate a correlation for a given trait. In this case **0.99**

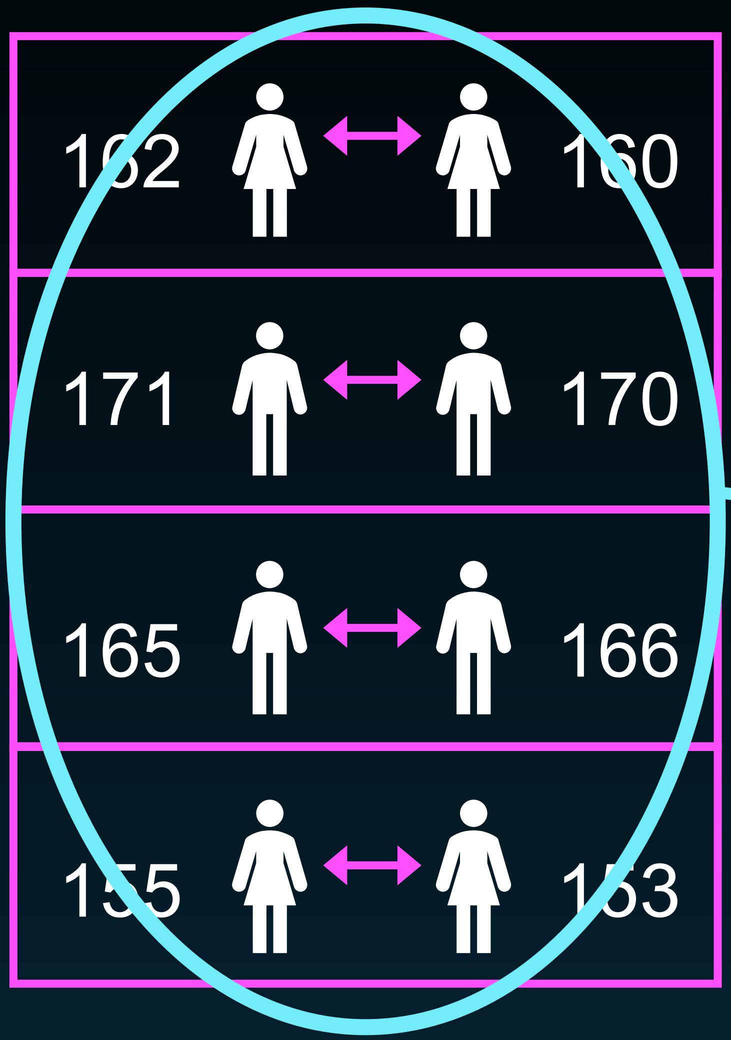
Monozygotic twins
(identical twins)



To measure the extent of resemblance in intelligence of monozygotic twins:

> One compares the variation of IQ scores within a twin pair

Monozygotic twins
(identical twins)



To measure the extent of resemblance in intelligence of monozygotic twins:

> One compares the variation of IQ scores within a twin pair

> And the variation of IQ scores between the twin pairs

Monozygotic twins (identical twins)

162	160
171	170
165	166
155	153

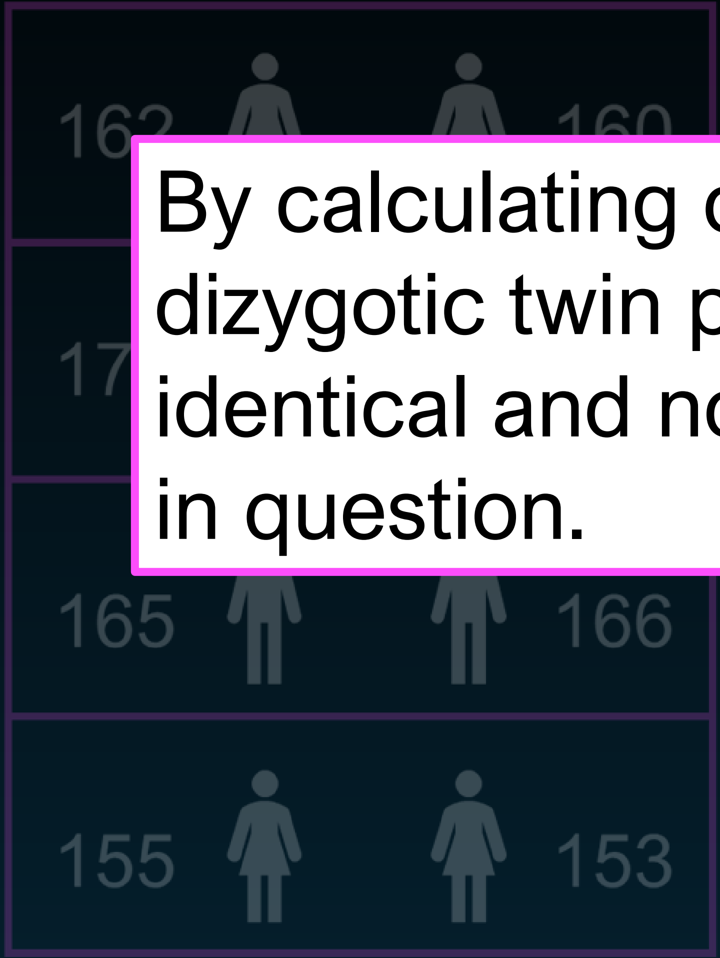
You can also conduct the same calculations for dizygotic twins

Dizygotic twins (non-identical twins)

173	178
165	170
174	164
158	163

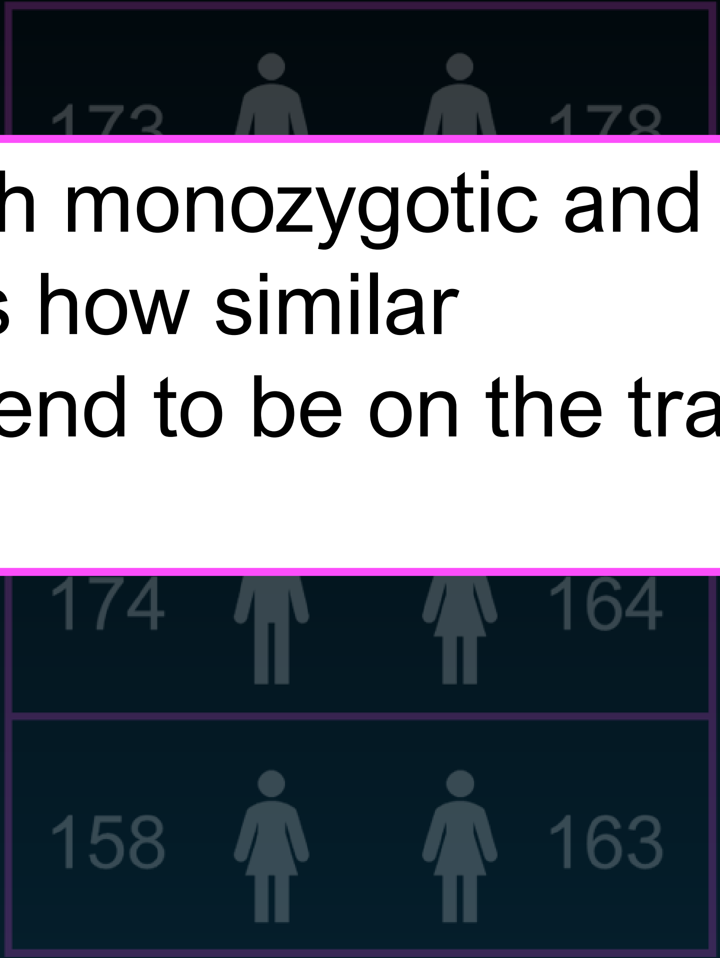
Monozygotic twins
(identical twins)

Dizygotic twins
(non-identical twins)



By calculating correlations for both monozygotic and dizygotic twin pairs: this will tell us how similar identical and non-identical twins tend to be on the trait in question.

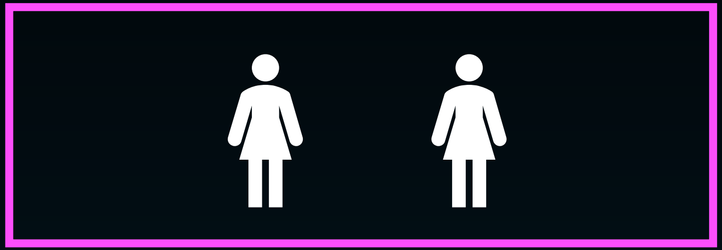
dizygotic twins



Summary:

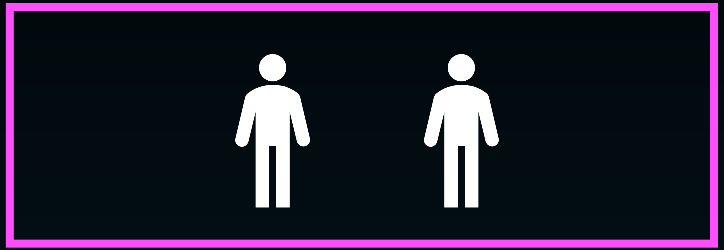
- Monozygotic twins share 100% of genes, and dizygotic share only 50% of genes
- It is assumed shared environment is just as similar between monozygotic and dizygotic twins
- Non-shared environment completely different within twin-pairs for both monozygotic and dizygotic twins
- In a nutshell, if identical twins are more similar to each other on intelligence test scores than non-identical twins, we can infer that genes play a role

Monozygotic twins
(identical twins)



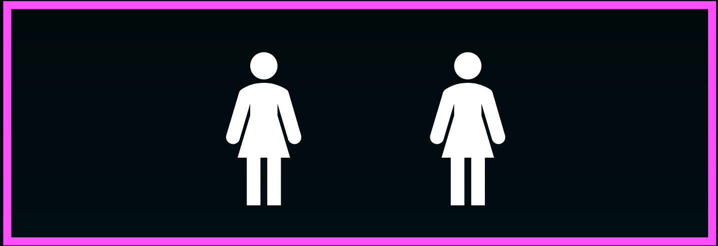
100% genes
+
Identical shared
Environment (100%)

Dizygotic twins
(non-identical twins)



50% genes
+
Identical shared
Environment (100%)

Monozygotic twins
(identical twins)



100% genes
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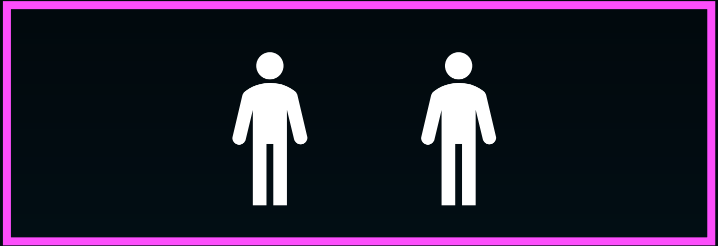
Identical shared
Environment (100%)

Dizygotic twins
(non-identical twins)



50% genes
+
Identical shared
Environment (100%)

Shared environment assumed to make the same contribution in both monozygotic & dizygotic sets of twins

Monozygotic twins
(identical twins)

Dizygotic twins
(non-identical twins)

100% genes

50% genes

Assumed that genetic effect similarity has twice as much effect in monozygotic twins (therefore reported differences multiplied by **2**)

Twins from around the world

Claire Haworth

- Dataset from multiple twin studies
- 4809 monozygotic twin pairs and 5880 dizygotic twin pairs
- Twins came from:
  > Ohio USA (mean age 5)
  > UK (mean age 12 years)
  > Minnesota USA (mean age 13 years)
  > Colorado USA (mean age 13 years)
  > Australia (mean age 16 years)
  > Netherlands (mean age 16 years)

Twins from around the world

Claire Haworth

- Largest sample was from UK (*Twins Early Development Study*)
- 1518 pairs of monozygotic twins; 2500 pairs of dizygotic twins (1293 same sex; 1207 opposite sex)
- The twin type was determined from Q.aire or if any doubt via a DNA test
- Four web-based cognitive tests. Three from WAIS for children and one Raven’s Matrices
- General intelligence score calculated

Twins from around the world

GHCA site	MZ	DZ
US Ohio	0.76 (0.68–0.83) (n = 121)	0.55 (0.44–0.65) (n = 171)
United Kingdom	0.67 (0.64–0.69) (n = 1518)	0.43 (0.40–0.46) (n = 2500)
US Minnesota	0.76 (0.73–0.78) (n = 1177)	0.50 (0.44–0.55) (n = 679)
US Colorado	0.82 (0.80–0.84) (n = 1288)	0.53 (0.49–0.56) (n = 1559)
Australia	0.83 (0.79–0.86) (n = 338)	0.48 (0.41–0.54) (n = 513)
Netherlands	0.83 (0.80–0.86) (n = 434)	0.58 (0.52–0.63) (n = 517)

Mean = 0.78

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Mean = 0.78 0.51

- Difference between twin types:
  = 0.78 – 0.51 = **0.27**
- Therefore **54%** of the differences in intelligence caused by genetic differences (remember twice the difference between MZ and DZ correlations, i.e. 0.27 x **2** = 0.54
- **24%** of differences caused by shared environment (MZ correlation of 0.78 minus the genetic effect of 0.54)
- **22%** of differences caused by non-shared environment (1 minus MZ correlation of 0.78)

Twins from around the world

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Mean = 0.78 0.51

- Roughly speaking for this childhood to young adulthood combined sample:
  - > About half the differences in intelligence are caused by genetic differences
  - > Half by environment

Twins from around the world

Age category	MZ	DZ
Childhood mean = 9yrs	0.74 (0.71–0.77) n = 1089	0.53 (0.49–0.57) n = 1591
Adolescence mean = 12yrs	0.73 (0.70–0.74) n = 2222	0.46 (0.43–0.49) n = 2712
Young adulthood mean = 17yrs	0.82 (0.80–0.83) n = 1498	0.48 (0.44–0.51) n = 1577

= 0.74 – 0.53 = **0.21**

= 0.73 – 0.46 = **0.27**

= 0.82 – 0.48 = **0.34**

Double the difference according to double the genetic effect in MZ vs DZ

- Combined sample was then separated by age

Twins from around the world

Age category	MZ	DZ
Childhood mean = 9yrs	0.74 (0.71–0.77) $n = 1089$	0.53 (0.49–0.57) $n = 1591$
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Young adulthood mean = 17yrs	0.82 (0.80–0.83) $n = 1498$	0.48 (0.44–0.51) $n = 1577$

0.21 x 2 → **42%**

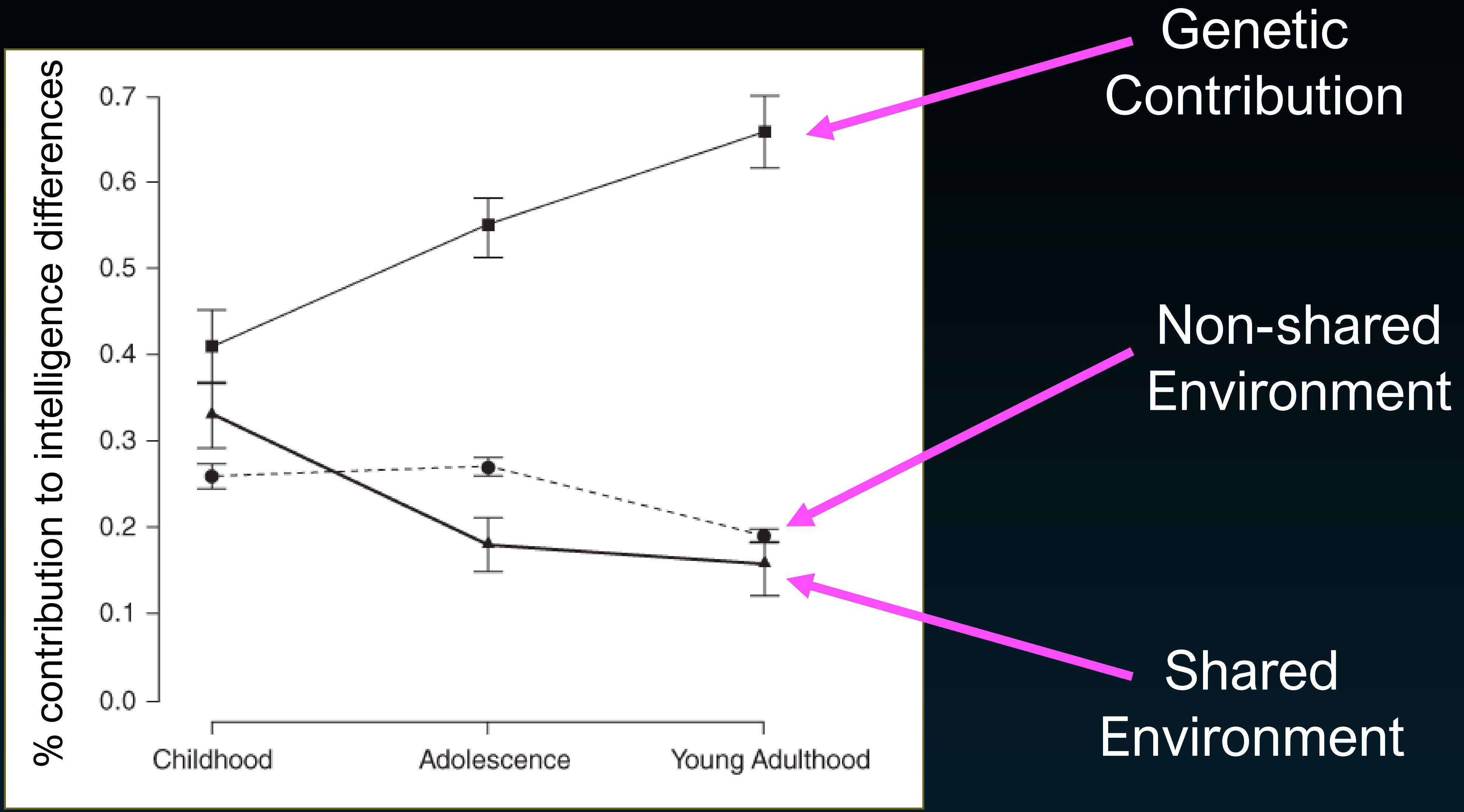
0.27 x 2 → **54%**

0.34 x 2 → **68%**

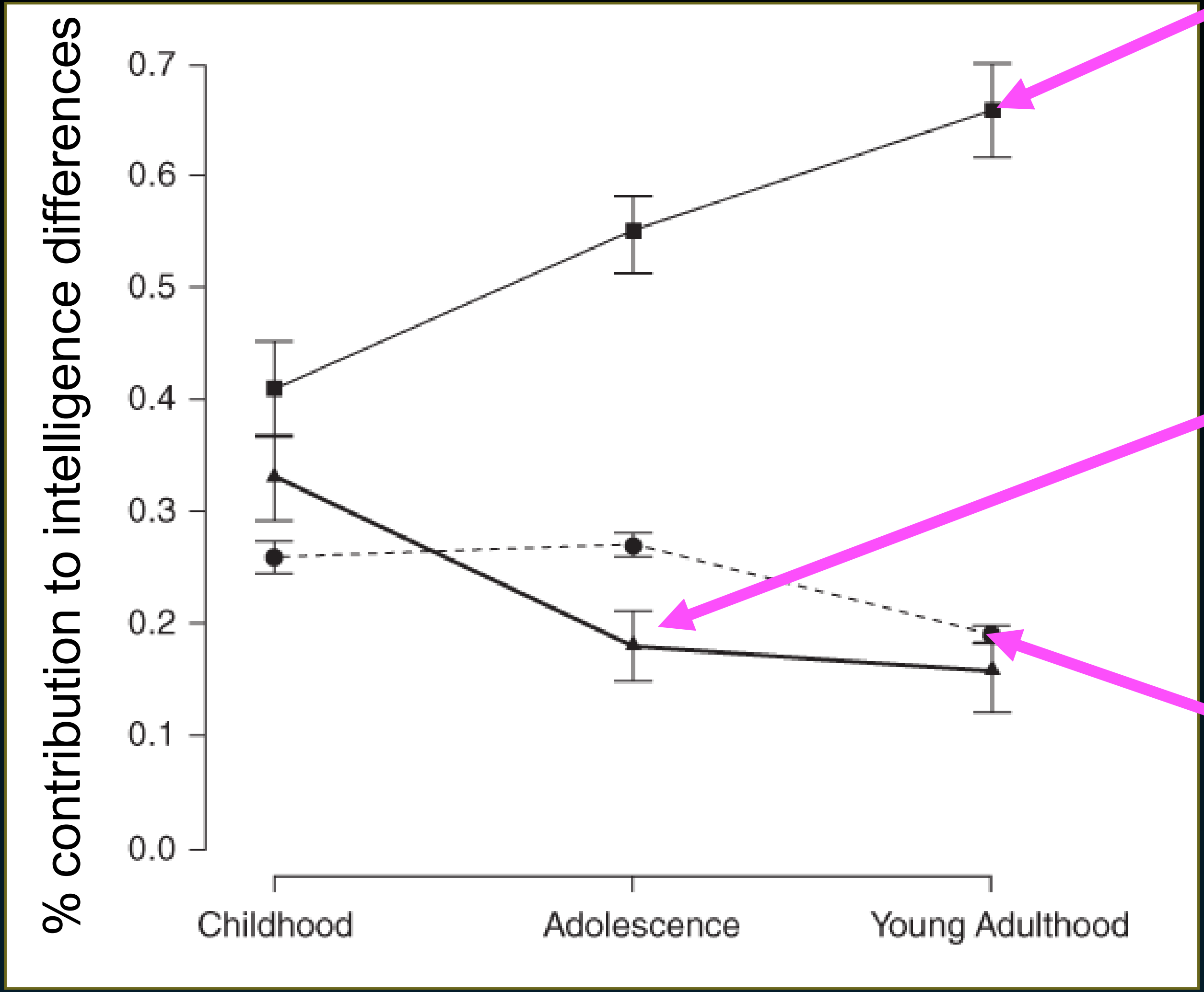
- Combined sample was then separated by age

% of intelligence differences caused by genetic factors

Twins from around the world

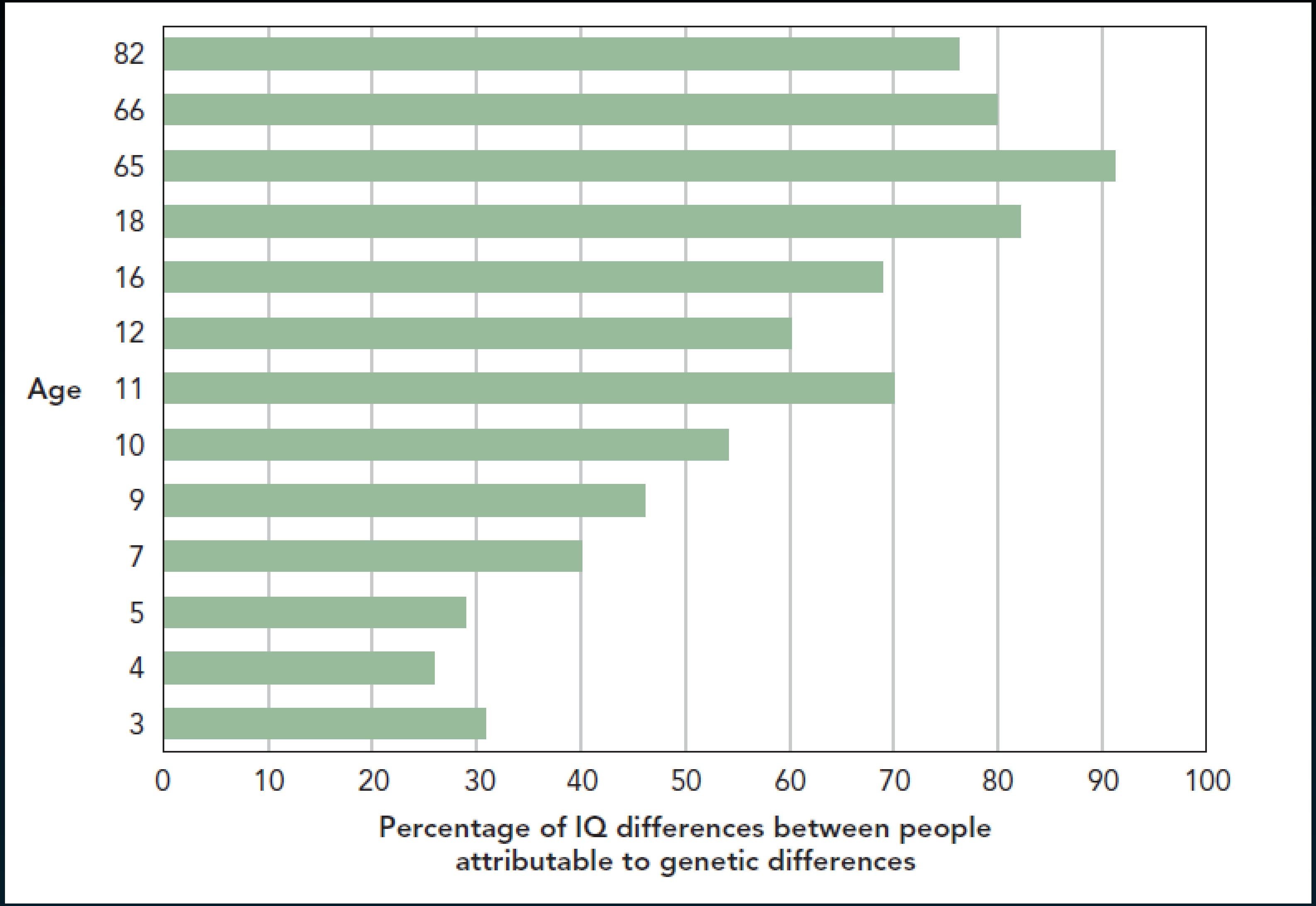


Twins from around the world



- Genetic contribution to people's differences in intelligence is larger in adulthood (66%) than in childhood (41%) and larger in adolescence (55%) than childhood
- Contribution of shared environment declines from 33% in childhood to 18% in adolescence and to 16% in adulthood
- Contribution of non-shared environment remains quite stable (around a 1/4th or 1/5th across the age range)

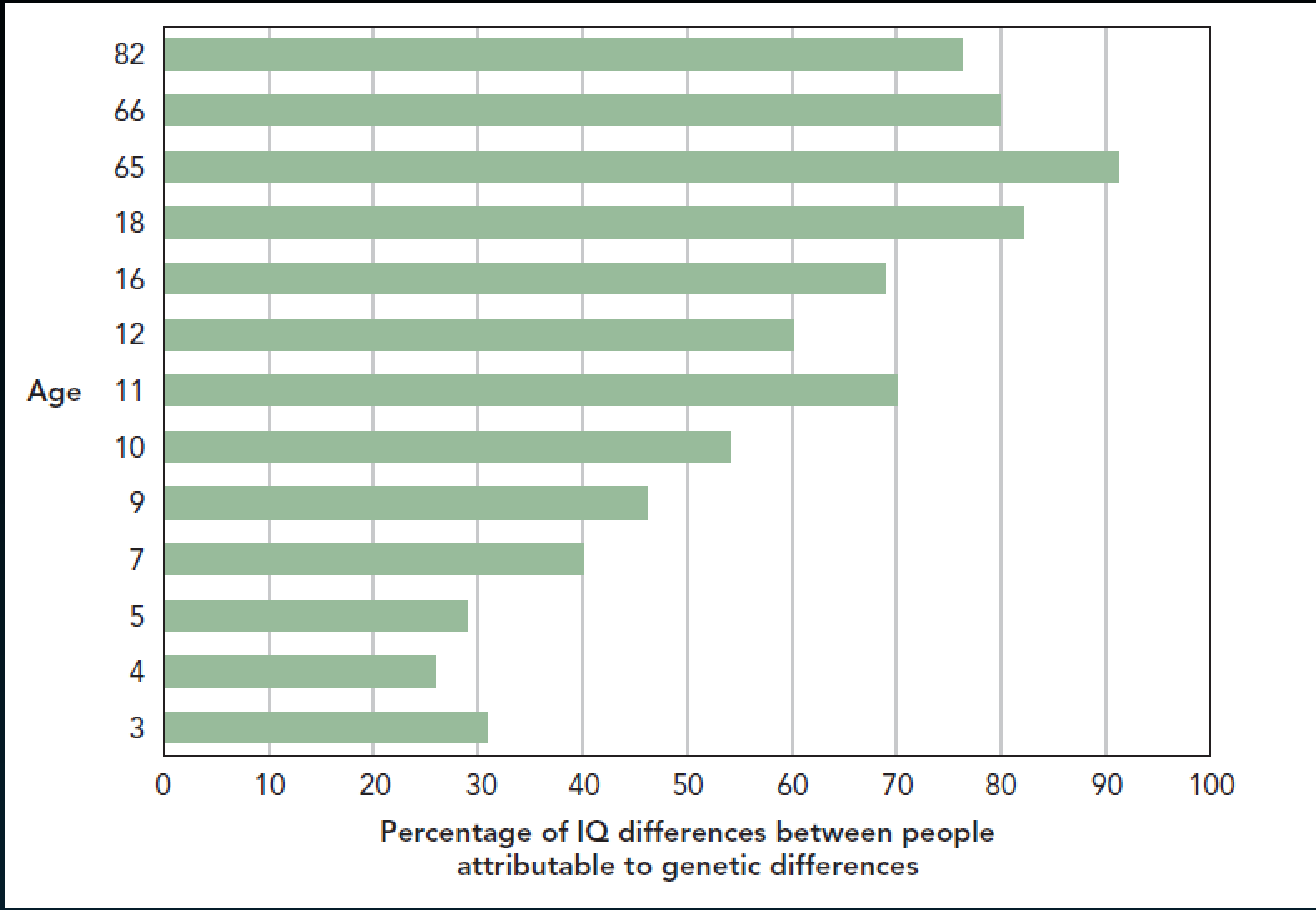
Heritability of intelligence increase with age



Taken from Schacter et al 2015

- Most genetic research been consistent: for intelligence, heritability increases linearly from childhood through adulthood
- Some evidence that heritability might increase to as much as 80% in later adulthood (Panizzon et al. 2014) but then decline after 80 years (Lee et al. 2010)
- Why?

Heritability of intelligence increase with age

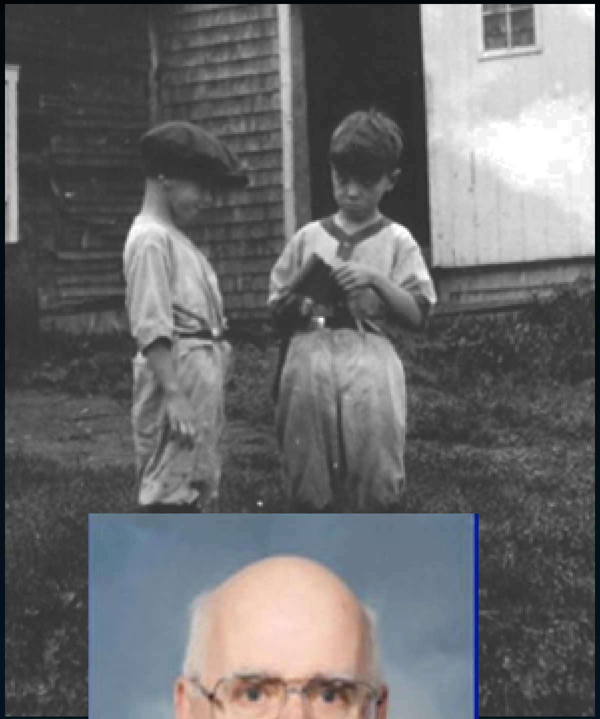


- One theory is that environments are more similar for any given pair of 65 year olds than a pair of 4 year olds so any differences in older people must be due to their genes
- Another theory is that as children grow up, they increasingly select, modify and even create their experiences, which are based on their genetic propensities

Taken from Schacter et al 2015

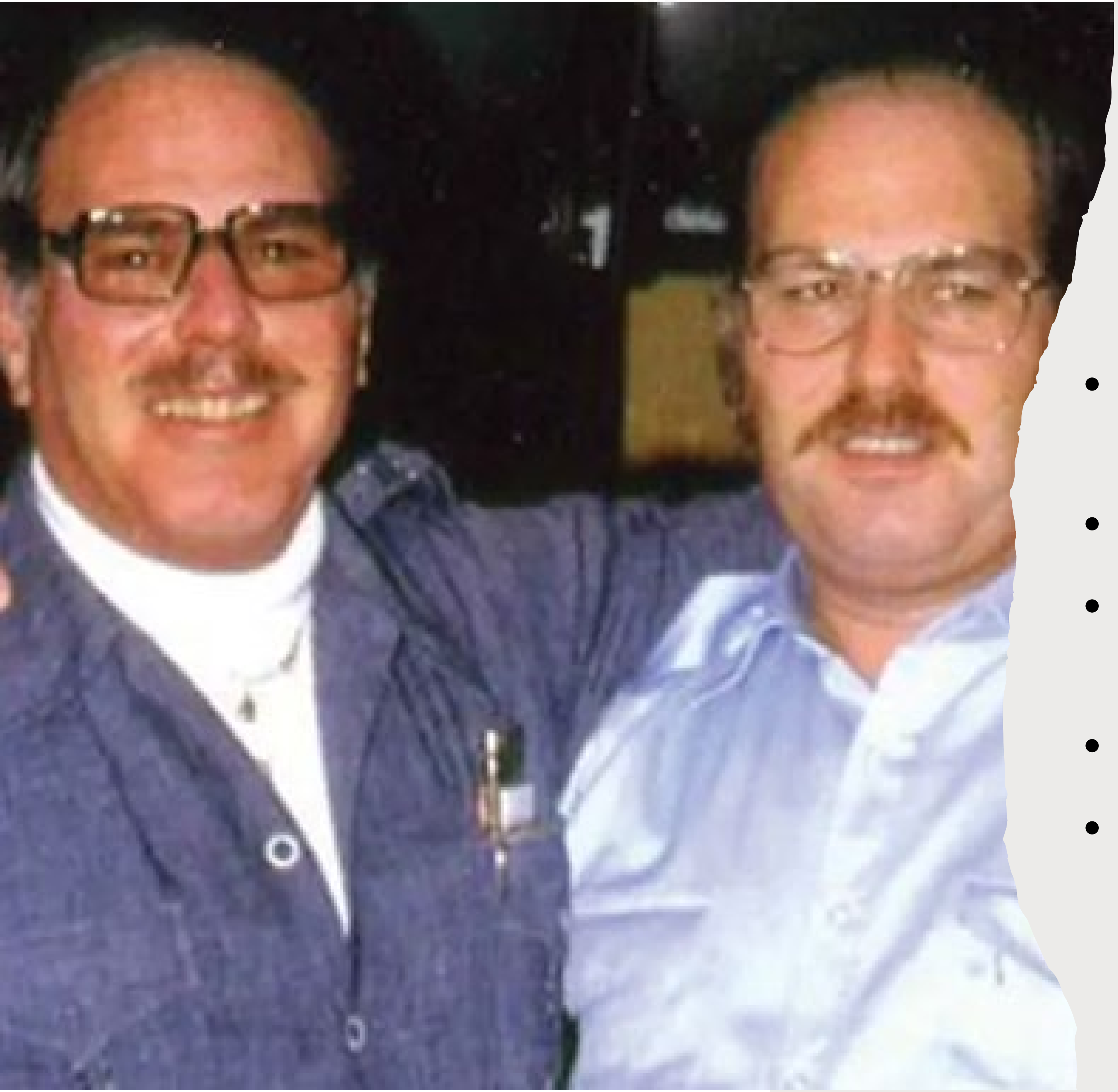
Minnesota Study of Twins reared apart (*MISTRA*)

Minnesota Study of Twins reared apart (*MISTRA*)



Thomas Bouchard

- 1979 Bouchard and colleagues led worldwide effort to find as many twins reared apart
- By the time it was terminated in 2000, they had enrolled 139 twin pairs
- Extensive intelligence testing
- Looking for genetic cascade whereby within correlations are higher for:
  - > MZ twins….then
  - > Lower for DZ twins….then
  - > Lower for distant realtives (e.g. cousins)
  - > And lowest for unrelated ppl. (e.g. classmates)



Minnesota Study of Twins reared apart (*MISTRA*)

- Jack and Oskar – separated six months after birth
- Jack raised as a jew in Trinidad
- Oskar grew up in Nazi Germany where he joined the Hitler youth
- 1999 BBC documentary
- These twins and others studied extensively

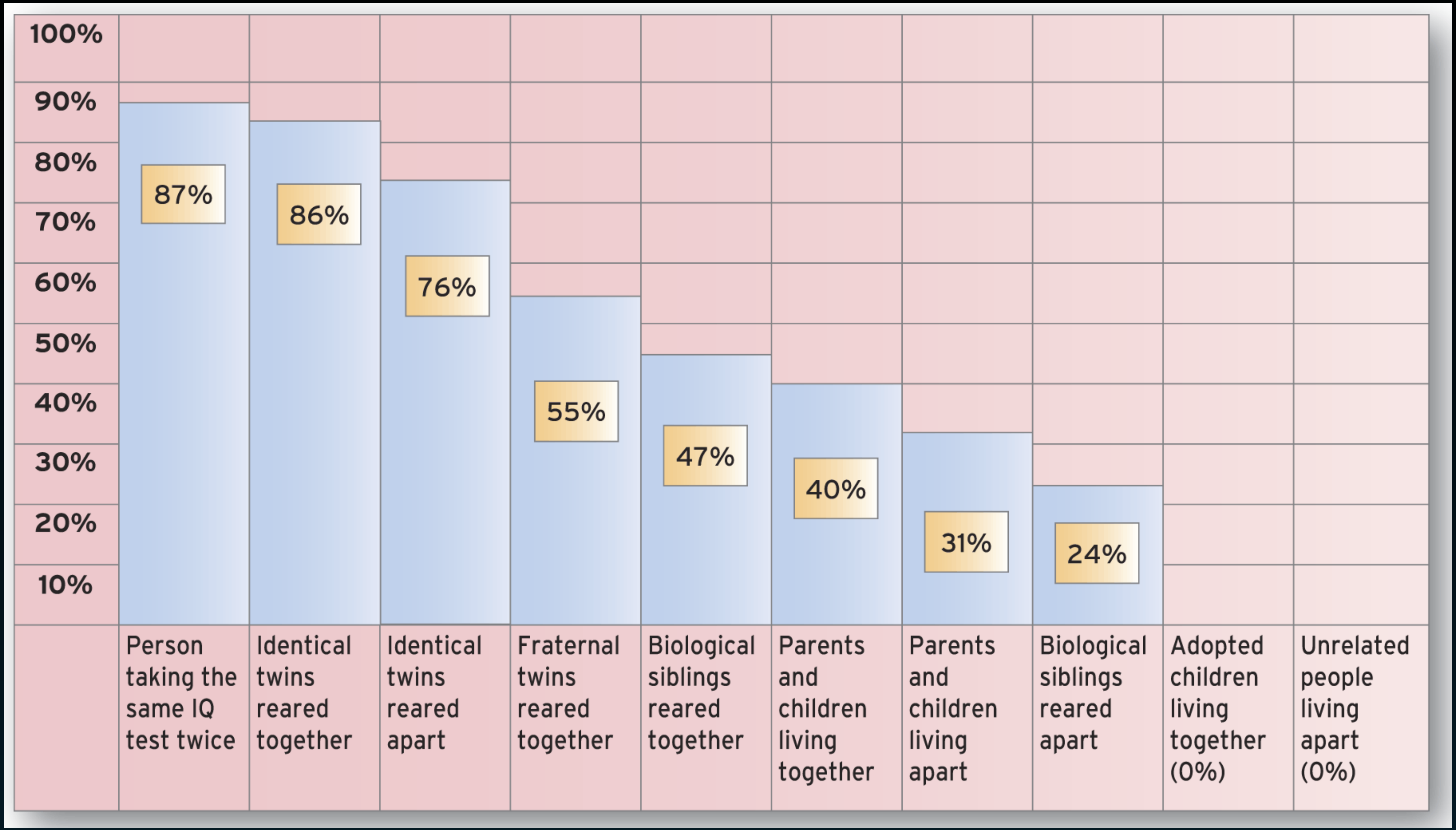


Figure 14.1 Concordance rates of intelligence

Source: From Ridley, Matt. (1999). *Genome: the autobiography of a species in 23 chapters.* London: Fourth Estate. Copyright © M. Ridley 1999. Reprinted with permission of HarperCollins Publishers Ltd

Problems with twin and adoption studies

- Both twins and adopted people are important for nature vs nurture comparisons but they may not be representative of normal populations
- Twin studies might over-estimate the role of genetics, e.g. identical twins may be treated more similarly by their parents, spend more time together and more often have the same friends. These environmental factors may drive similarities beyond genetics.
- Adoption strategies might also influence heritability. For example, an adopted child would rarely be placed into a household suffering from poverty or abuse

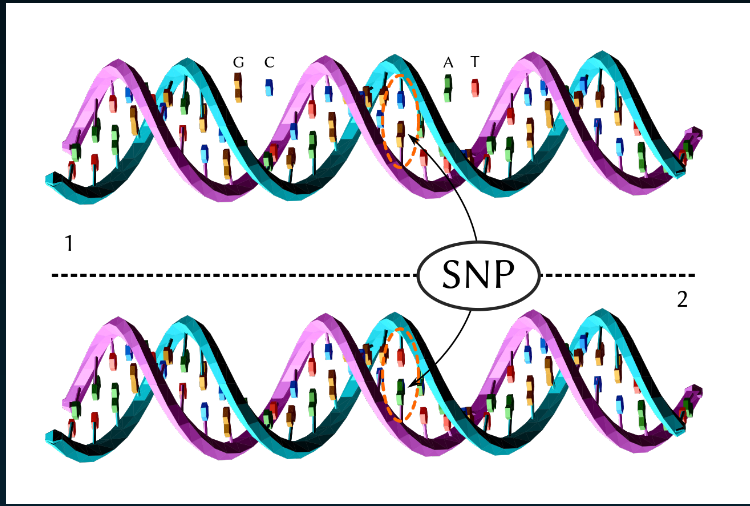
DNA studies

Looking at DNA

- Genome-wide association (GWAS) studies
- Human genome estimated to comprise of 3 billion nucleotides of DNA within 23 chromosome pairs
- 3 billion stretch has some structure
- Genes – stretches of DNA coding for proteins
- About 19,000 genes in human DNA
- Genes code for proteins (made of amino acids)
- Across the 3 billion stretch there are differences between ppl. At any one site, there is the commonest nucleotide for a given population.
- Ppl. have a different nucleotide at about 1 in every thousand places, so a person likely to have 4-5 million places in DNA differing from the most usual nucleotide (called single nucleotide polymorphisms - SNPs)

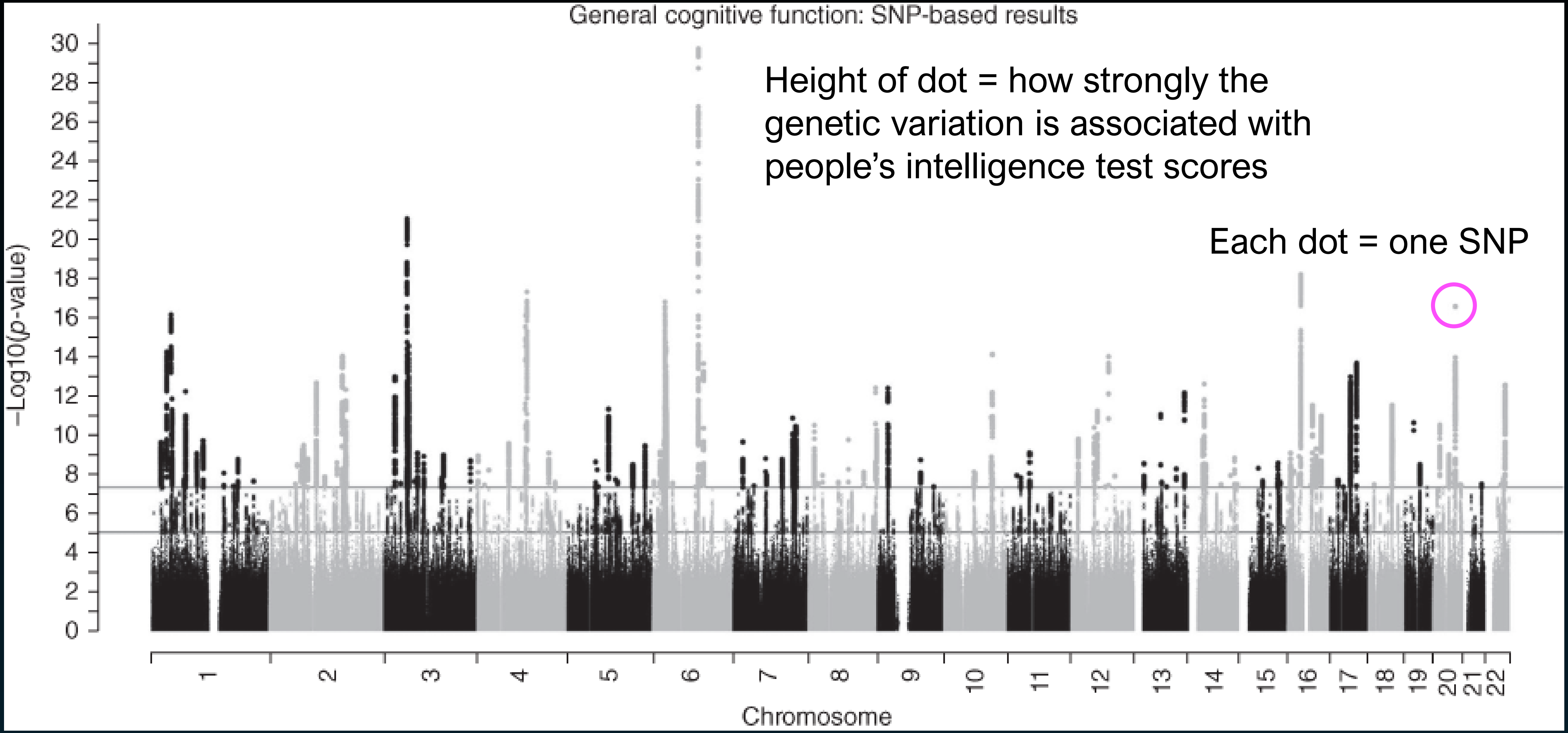
Looking at DNA

- How does having one single nucleotide polymorphism (SNP) vs another at a given position on a chromosome affect humans?
- Examined intelligence and DNA SNPs of 300,486 ppl. from 57 studies around the world.
- From battery of intelligence tests, a *g* score was calculated
- Study reported on almost 13 million SNPs



Davies et al. (2018) *Nature Communications*

Studies looking at DNA



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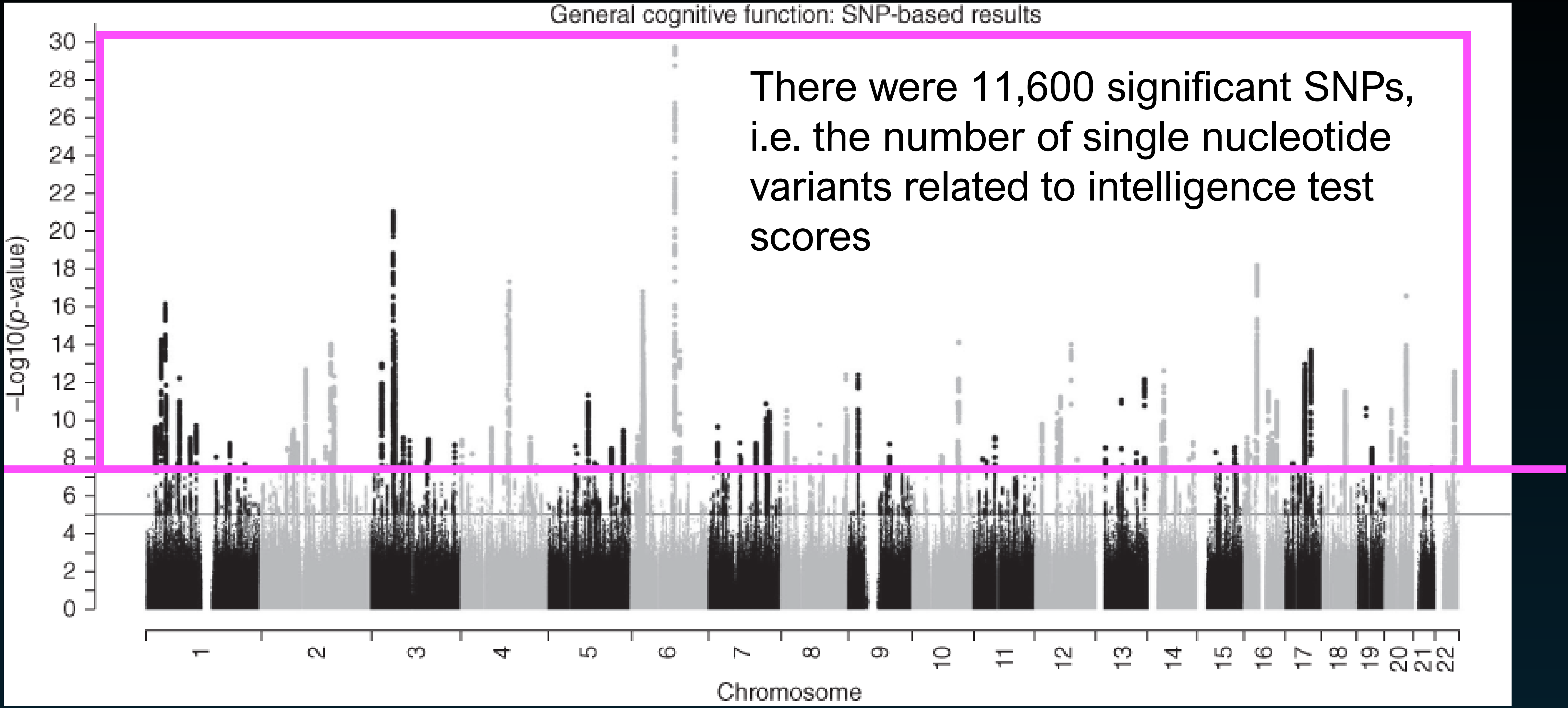


Davies et al. (2018) *Nature Communications*

Studies looking at DNA

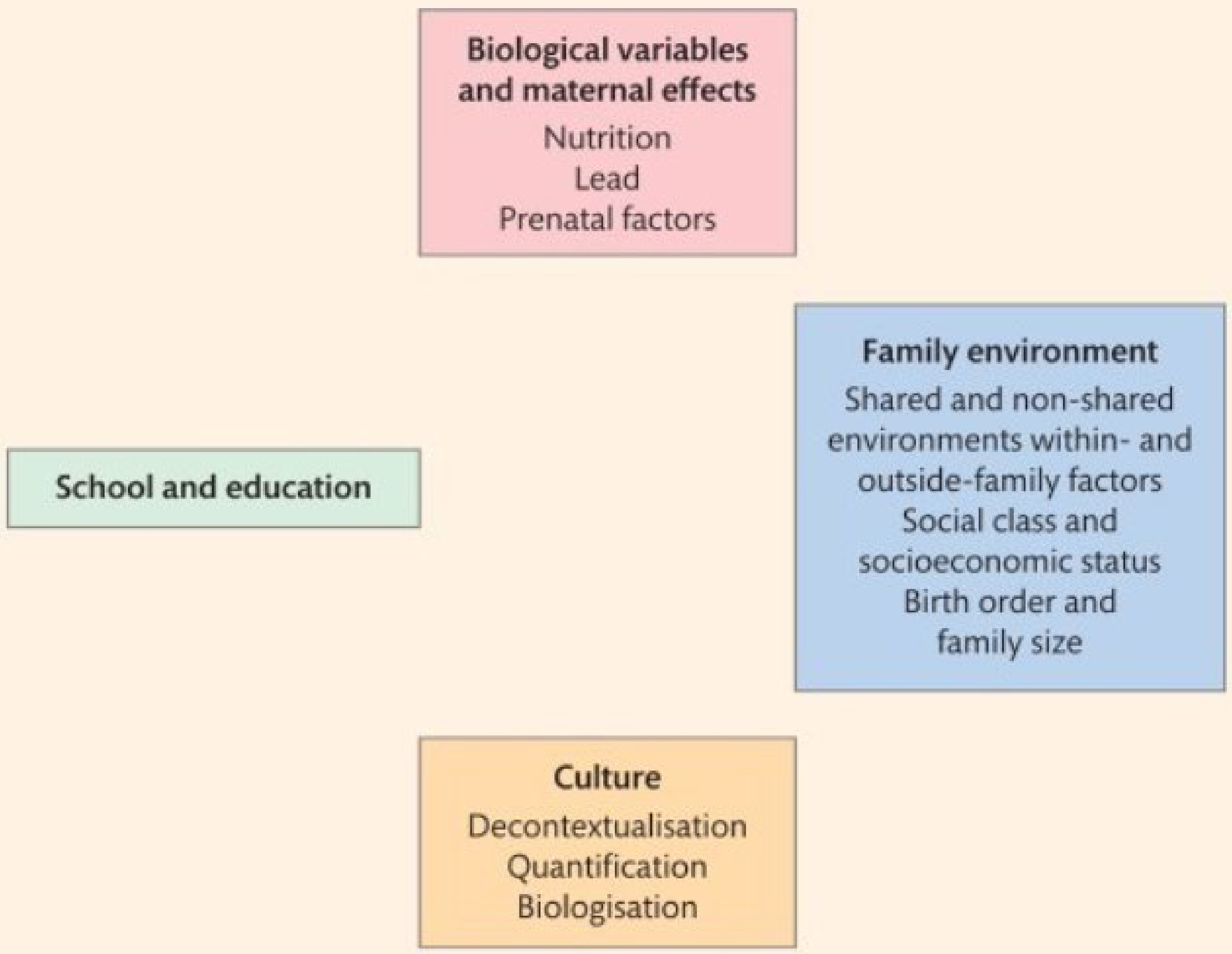
What has been learned from this GWAS study?

- Large numbers of genetic variants in a large number of genes related to people's intelligence scores
- The heritability of intelligence using this method (25%) is about half of that found using twin studies – gap should shorten with more genetic variation studies
- Using people's DNA, one can predict better than chance their intelligence differences - currently prediction not strong but should get better with more and more data
- Still early days with other genetic variants other than SNPs to be studied
- Also found some of the genetic variants related to intelligence also related to body and health traits including brain volume and longevity

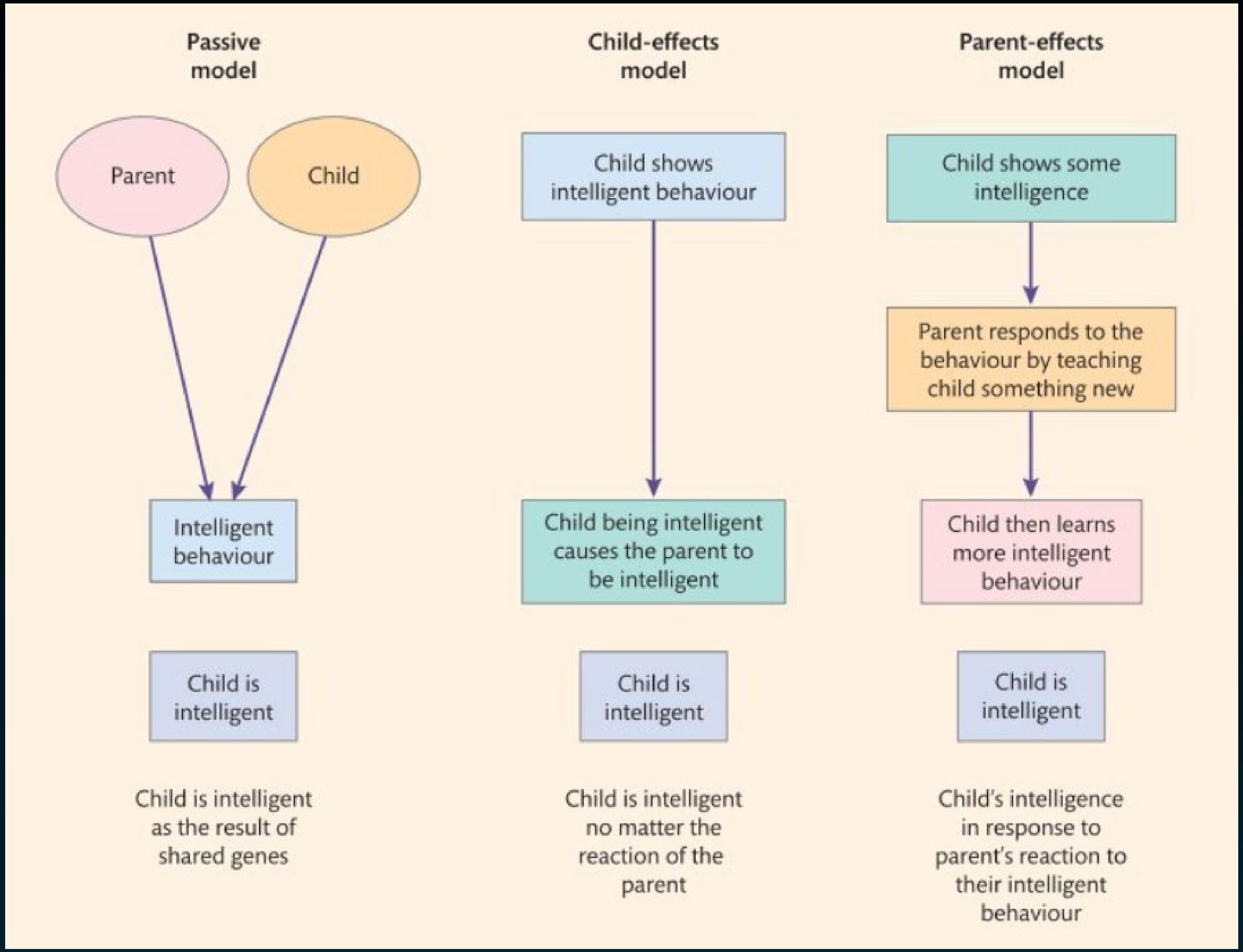
Davies et al. (2018) *Nature Communications*

Environmental influences on intelligence

American Psychological Association Task Force (Neisser et al., 1996)



Within-family factors



Stability of intelligence across lifetime

Stability of intelligence across our lifetime?

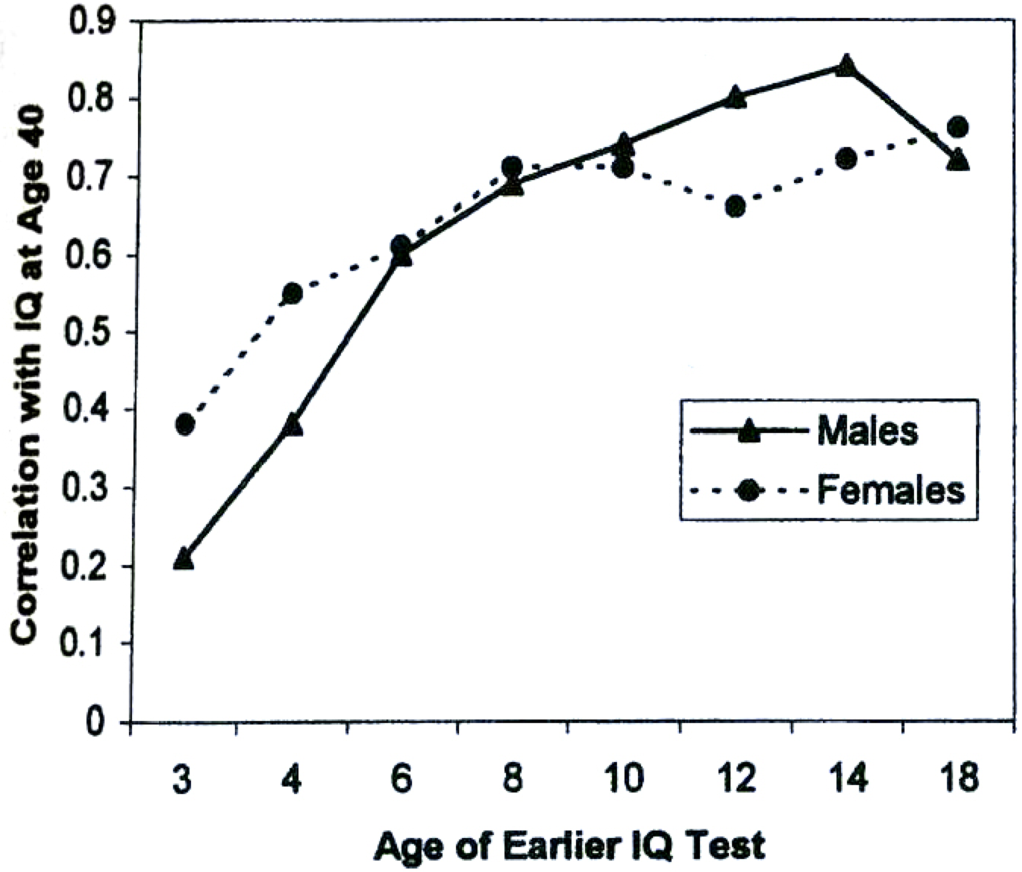
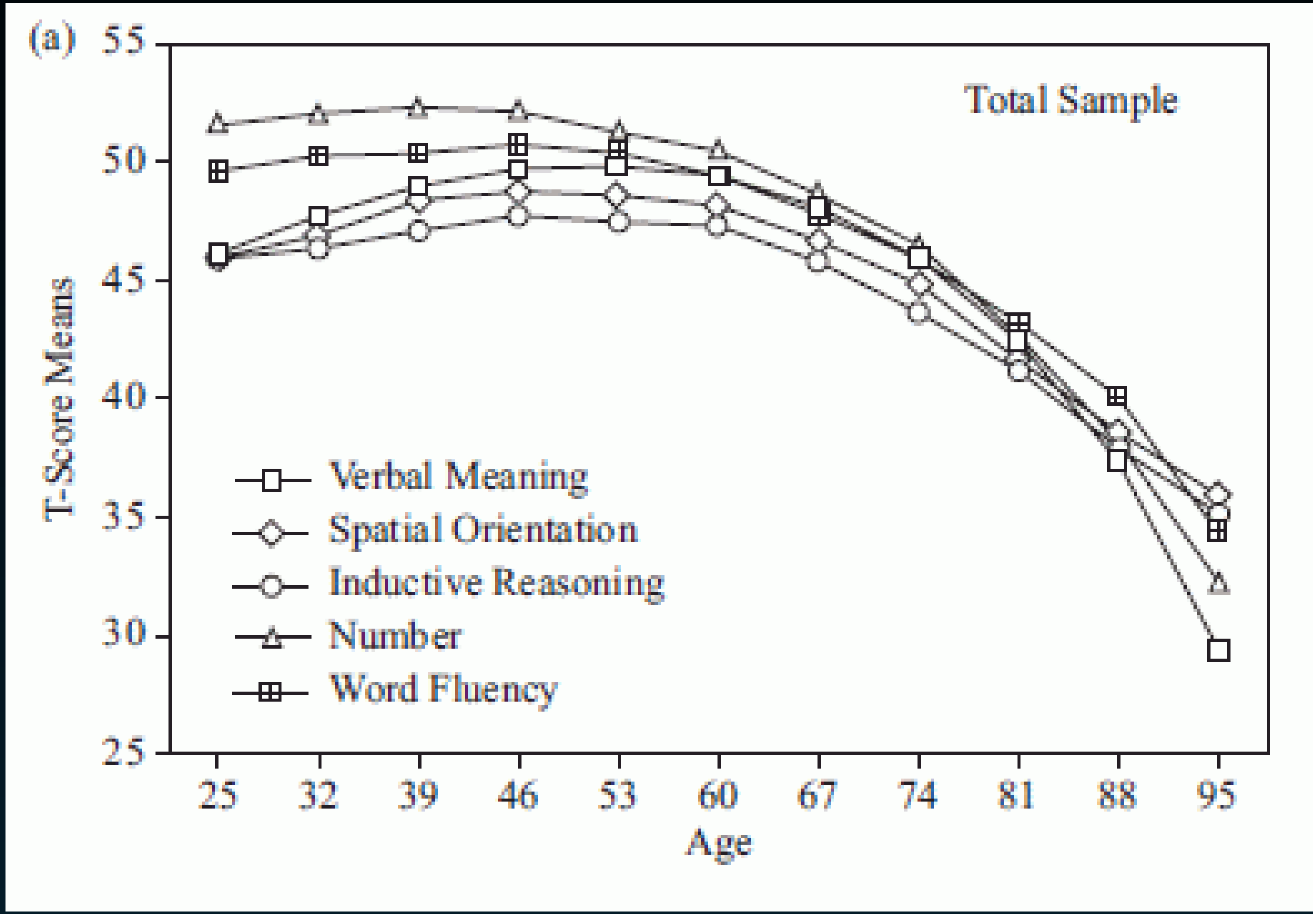


Fig. 2.5 The long-term stability of IQ scores in males and females. The ordinate of the figure shows the value of the correlation between earlier IQ and IQ at age 40, and the abscissa shows the age at which the earlier IQ score was measured. (Data from a study by Honzik, reported by McCall, 1977.)

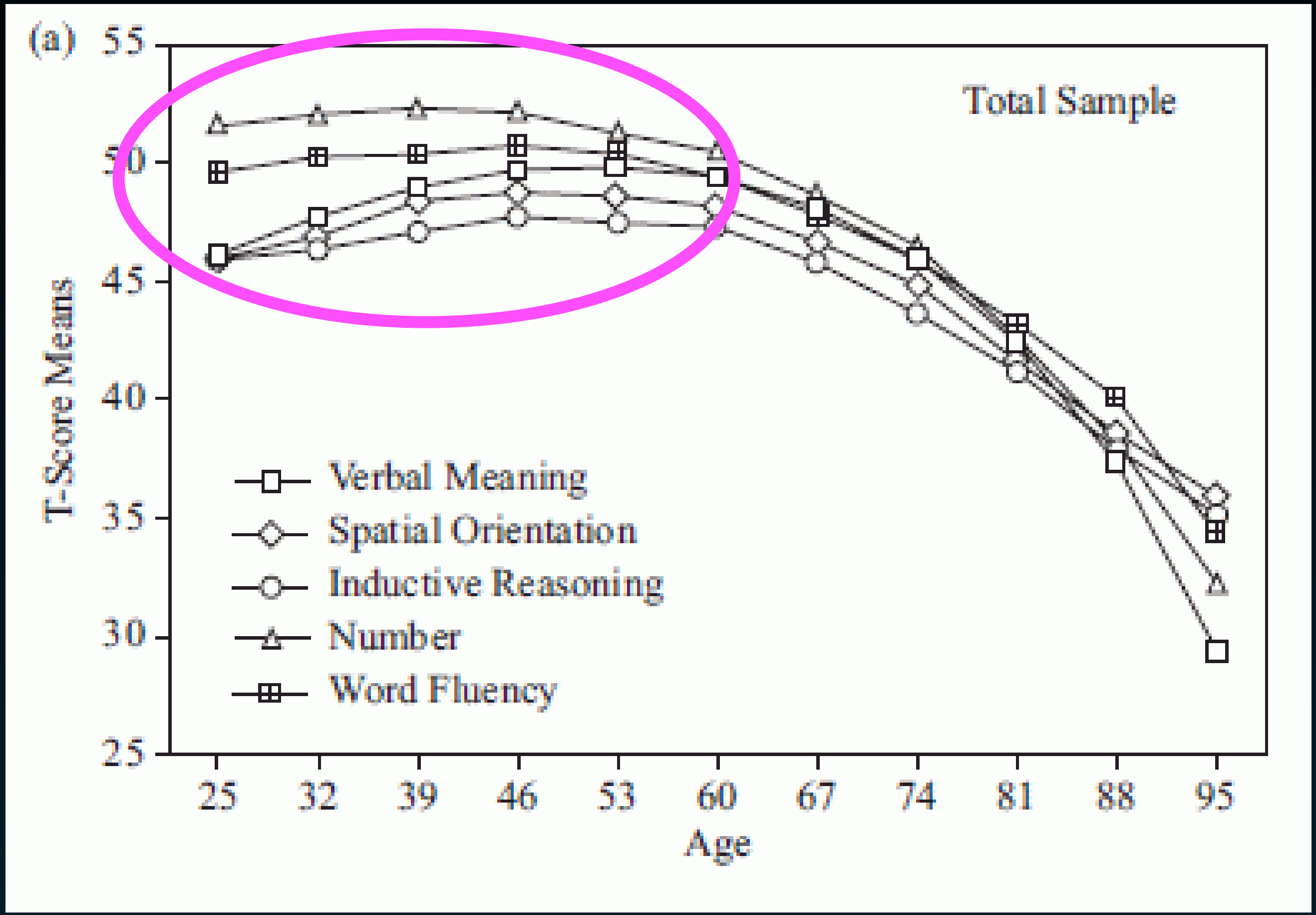
Stability of intelligence across our lifetime?



Schaie, KW (2015)

Stability of intelligence across our lifetime?

Cognitive abilities remain relatively stable from 25 to 65 years of age



Schaie, KW (2015)

Are generations getting smarter?

The Flynn effect

- The scores on intelligence tests have a tendency to fluctuate.
- The scores of intelligence tests change continuously from year to year.
- Flynn (1981) discovered a year-on-year rise of intelligence test scores (whenever a new/revised IQ test was compared to an older/previous IQ test).
- In particular, Flynn found that if a group of participants averaged an IQ of 100 on the original WISC (1947/1948), they averaged 108 on the WISC-R (1972).

→ **in a period of 24 years the IQ gain from WISC to WISC-R was ~8 IQ points**

- Flynn (1984) looked at 73 studies (covering 7,500 US white participants aged between 2 and 48 years).

Country, the Test and Type of Test (verbal or non-verbal or both) Administered, the Growth in IQ Points Across Many of the Samples and a Figure of Average Growth in IQ per Year

Location	Test	Type of IQ test (verbal or non-verbal)	Period of study	Rise in IQ points over the period	Rate
Leipzig, East Germany	Raven's	non-verbal	1968–1978	10.00–15.00	1.250
France	Raven's	non-verbal	1949–1974	25.12	1.005
Japan	Wechsler	non-verbal/verbal	1951–1975	20.03	0.835
Vienna, Austria	Wechsler	non-verbal/verbal	1962–1979	12.00–16.00	0.824
Belgium	Raven's	non-verbal	1958–1967	7.15	0.794
West Germany	Wechsler	non-verbal/verbal	1954–1981	20.00	0.741
Belgium	Shapes	non-verbal	1958–1967	6.45	0.716
Netherlands	Raven's	non-verbal	1952–1982	20.00	0.667
Zurich, Switzerland	Wechsler	non-verbal/verbal	1954–1977	10.00–20.00	0.652
Norway	Matrices	non-verbal	1954–1968	8.80	0.629
Saskatchewan, Canada	Otis	verbal	1958–1978	12.55	0.628
West Germany	Horn-Raven's	non-verbal	1961–1978	10.00	0.588
Norway	Verbal-Math Test	verbal	1954–1968	8.15	0.582
Edmonton, Alberta, Canada	California Test of Mental Maturity	non-verbal/verbal	1956–1977	11.03	0.525

(Flynn, 1987)

Country, the Test and Type of Test (verbal or non-verbal or both) Administered, the Growth in IQ Points Across Many of the Samples and a Figure of Average Growth in IQ per Year

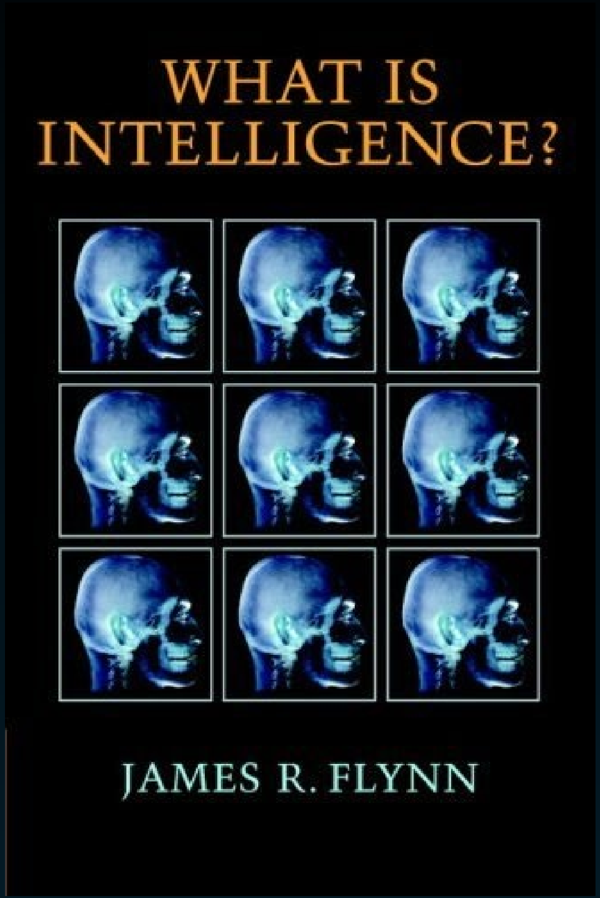
Location	Test	Type of IQ test (verbal or non-verbal)	Period of study	Rise in IQ points over the period	Rate
Australia	Jenkins	non-verbal	1949–1981	15.67	0.490
Belgium	Verbal-Math Test	verbal	1958–1967	3.67	0.408
Edmonton, Alberta, Canada	Raven's	non-verbal	1956–1977	8.44	0.402
France	Wechsler	non-verbal/verbal	1955–1979	9.12	0.380
France	Verbal-Math	verbal	1949–1974	9.35	0.374
Saskatchewan, Canada	Otis	verbal	1958–1978	6.95	0.348
Australia	Raven's	non-verbal	1950–1976	8.76	0.337
United States	Wechsler-Binet	non-verbal/verbal	1932–1972	12.00	0.300
United States	Wechsler	non-verbal/verbal	1954–1978	5.95	0.243
New Zealand	Otis	verbal	1936–1968	7.73	0.242
Norway	Matrices	non-verbal	1968–1980	2.60	0.217
United Kingdom	Raven's	non-verbal	1938–1979	7.75	0.189
Solothurn, Switzerland	Wechsler	non-verbal/verbal	1977–1984	1.30	0.186
United Kingdom	Raven's	non-verbal	1940–1979	7.07	0.181
Norway	Verbal-Math	verbal	1968–1980	−1.60	−0.133

(Flynn, 1987)

Results of Flynn‘s (1987) meta analysis

- The highest rise in IQ occurred in the non-verbal tests (fluid intelligence; 5.9 IQ points per decade) and lowest gains were in verbal tests (crystallised intelligence; 3.7 IQ points per decade)

→ Due to ***better schooling***? (However, IQ gains in verbal vs. non-verbal IQ should be reversed)



Cambridge University Press

Explanations of the Flynn effect

- ***Generations are getting more and more intelligent***

→ unlikely, we would expect to find more geniuses in the world

→ these gains are far too rapid to result from genetic changes

- ***Length of schooling***

→ important, but mainly relevant for verbal tests

- ***Test-taking sophistication*** (people become more familiar with IQ tests)

→ does not explain the difference between verbal and non-verbal IQ tests

- ***Child-rearing practices*** (parents are more interested in their children‘s intellectual development)

→ some evidence, but educational programs have probably no lasting effects on IQ

- ***Cognitive stimulation hypothesis*** (visual and technical environment, TV learning)

→ little direct evidence

- ***Nutrition*** (e.g. nutrition has been linked to brain size)

SUMMARY

- Heritability of intelligence has been studied using both twin studies and with more recent DNA (GWAS) techniques
- Pattern of correlations from lots twin study data combined suggests about 50% of intelligence various is due to genetic factors
- Combination of twin studies tells us that genes and environments matter for people's intelligence differences
- Genetic variation goes up from childhood to adulthood – while the effects of shared environments decrease